

SINGLE-SOURCE RESEARCH REPORT

Surreal and surcomplex difference fields

A Single Dilation Recovers Hahn Support

Exact difference cohomology, definable normal forms,
and centralizers of monomial automorphisms

Research article prepared with ChatGPT
for Vladimir Reshetnikov

22 September 2026

Built from one manuscript of batch 22 of the collection, pinned to repository commit 465a54b

Abstract

Let $K = \mathbf{k}((t^\Gamma))$ be a full Hahn field of characteristic zero. We construct a strong, valuation-nondecreasing partial inverse of $\sigma - \lambda$ for every coefficient-fixing monomial automorphism $\sigma(t^g) = \chi(g)t^{\theta g}$. The only obstructions occur at exponents fixed by θ on which χ equals λ . For finitely many commuting such automorphisms, these inverses give an explicit deformation retraction of the entire Koszul complex onto its joint-resonant coefficients. This computes all additive cohomology groups and the principal-unit part of multiplicative cohomology, and gives exact criteria for simultaneous multiplicative equations.

The main structural application is that the single automorphism $S_2(\sum a_g t^g) = \sum a_g t^{2g}$ defines the coefficient field, constant coefficient, canonical monomials, every coefficient, the valuation ring, and truncation in the language of fields with one automorphism. Consequently its centralizer among *all* field automorphisms consists precisely of untwisted coefficient–exponent lifts; neither valuation preservation nor strong additivity needs to be assumed. A uniform interpretation of full monadic second-order arithmetic yields undecidability, and a coefficient formula directly witnesses the tree property of the second kind. All nonidentity rational dilations give the same definable normal-form data when Γ is divisible, although distinct rational dilations are not conjugate. The elementwise results apply to **No** and **No**(i) through Conway normal forms, with proper-class qualifications stated explicitly.

Status. Complete proofs are supplied for the stated results. Novelty is a qualified research claim, not certified priority. The accompanying finite checks are not a formal verification. No solution to the published exponential-automorphism questions is asserted. An AI-assisted research draft, not refereed; no theorem of this report is Lean-verified. The exponent dilation is written S_q , as in the collection’s surcomplex-automorphisms report; the source wrote D_q (Section 1.4). Relations to the collection as it now stands are in Section 1.5 and Appendix D. Every label carries the prefix **dsup**∶.

Contents

1	Introduction and research position	3
1.1	Principal results	3
1.2	What is inherited and what is proposed as new	4
1.3	Repository snapshot and evidence boundary	4
1.4	Notation and conventions in this report	5
1.5	Relation to other reports in the collection	6
2	Conventions, supports, and monomial automorphisms	7
3	The orbit-support lemma	8
4	An exact resolvent and its resonance obstruction	9
5	Commuting equations and an explicit Koszul contraction	11
6	Principal units and multiplicative difference equations	13
6.1	The formal logarithm is the appropriate logarithm	13
6.2	One multiplicative equation	14
6.3	A commuting multiplicative system	14
7	Rational dilations: only scalar obstructions survive	15
8	Doubling defines the entire coefficient structure	16
8.1	Constants and the constant-coefficient map	17
8.2	The equation that recognizes monomials	17
8.3	All coefficients and the support relation	17
8.4	Recovering the orientation of the valuation	18
8.5	The valuation ring, leading term, and truncation	18
9	The full centralizer of doubling	19
10	Arithmetic interpreted by one automorphism	20
10.1	A uniform Boolean-series interpretation	20
10.2	A direct coefficient witness to TP_2	21
11	Every nonidentity rational dilation recovers the same data	21
12	Arbitrary-rank examples and sharp distinctions	22
12.1	A shear with a large fixed Hahn field	22
12.2	A phase character and an integral-exponent obstruction	23
12.3	Compatibility is not the same as exactness	23
13	Transfer to surreal and surcomplex normal forms	24
13.1	The normal-form dictionary	24
13.2	Set-sized localization of every proof	24
13.3	What is recovered, and what is not	25
14	A formalization route and precise limitations	26
14.1	A dependency-oriented route	26
14.2	Claims deliberately not made	26
14.3	Consolidated ledger of non-claims	26
15	Further questions suggested by the results	29
16	Conclusion	29
A	Endpoint identities and finite verification	30

B	A compact formula dictionary	30
C	Provenance, reproducibility, and novelty ledger	31
D	Provenance of this report	32
D.1	One manuscript	32
D.2	Corrections since the pin	33
D.3	Files as placed and rerun	33

1 Introduction and research position

A normal form is often treated as external information attached to a non-Archimedean field. The underlying field need not visibly identify its coefficient field, its chosen monomials, or the operation of deleting a part of a support. This article shows that one particularly simple field automorphism can recover all of this information.

The automorphism is exponent doubling. It does not square a general series:

$$S_2\left(\sum_g a_g t^g\right) = \sum_g a_g t^{2g}, \quad \left(\sum_g a_g t^g\right)^2 = \sum_{g,h} a_g a_h t^{g+h}.$$

The contrast between these operations is useful. The equation $S_2x = x^2$ recognizes precisely the canonical monomials. A second equation, $S_2y - y = x - c$, recognizes the constant coefficient c of x . Together these two equations recover arbitrary coefficients. Even the direction of the valuation is recovered from the constant coefficient of $(1 - m)^{-1}$ for a monomial m . This works over $\mathbb{C}$, without ordering the field or naming complex conjugation.

The difference equation used here admits a considerably more general solution. We develop that solution first, rather than relying on a special calculation for doubling. Its extension to commuting automorphisms gives an explicit cohomology calculation and separates multiplicative obstructions into a monomial part and a resonant logarithmic part.

1.1 Principal results

Theorems 4.1, 5.2, 8.1, and 9.1 are the main structural statements. Their content is as follows.

1. For $\sigma(t^g) = \chi(g)t^{\theta g}$ and $\lambda \neq 0$, restriction to $R_\lambda = \{g : \theta g = g, \chi(g) = \lambda\}$ is exactly both the kernel projection and the cokernel obstruction for $\sigma - \lambda$. A strong normalized inverse is given by explicit sums along the increasing halves of exponent orbits.
2. For a commuting d -tuple, let F be its common fixed field. There is an explicit strong contraction of the nonresonant part of the Koszul complex, and

$$H^r(\mathbb{Z}^d, K_{\text{add}}) \simeq F^{\binom{d}{r}}.$$

The analogous statement holds for the additive infinitesimal ideal; the Hahn logarithm turns it into a statement about principal units.

3. In (K, S_2) , for nonzero 2-divisible Γ , the entire canonical coefficient and truncation structure is parameter-free definable. In particular,

$$\text{Cen}_{\text{Aut}(K)}(S_2) \simeq \text{Aut}(\mathbf{k}) \times \text{Aut}_{\text{ord}}(\Gamma).$$

Every member of this centralizer is automatically strong.

4. The same expansion uniformly interprets $(\mathbb{N}, \mathcal{P}(\mathbb{N}); +, \in)$ and has an undecidable complete theory. A definable coefficient relation has TP_2 . Over divisible Γ , all nonidentity positive rational dilations are interdefinable, but they are pairwise nonconjugate as field automorphisms.

Here “strong” refers to preservation of Hahn summation, not convergence of finite partial sums. Every assertion about ordinary group cohomology is initially made for a set-sized Hahn field. Section 13 then states exactly what transfers to the proper-class surreal fields.

1.2 What is inherited and what is proposed as new

The construction of monomial automorphisms is established Hahn-field technology, not a contribution of this article. Kaplan–Krapp–Serra [1] study surreal automorphism groups using normal forms. The repository’s surcomplex-automorphism report [7] also develops coefficient, exponent, and character lifts. We use this construction but do not use its more ambitious classification assertions as premises.

The specialization $g \mapsto qg$ lies in Mahler difference algebra. Faverjon–Roques [3] treat algorithmic questions for general linear Mahler equations with Hahn-series solutions. Our first-order, constant-scalar resolvent does not replace their higher-order variable- coefficient theory. Its role here is an arbitrary-rank support theorem and an input to a simultaneous cohomology calculation.

Related (batch 34). Linear Mahler equations in a function variable, with arbitrary-rank Hahn coefficients, now appear in the holonomic-rigidity report (surcomplex/holonomic-rigidity-for-entire-hahn-functions). Its `hol:pc:cor:linear` shows that every strongly entire solution of a mixed differential–Mahler equation $\sum_j L_j(f(P_j(z))) = g(z)$ is a polynomial when the top pullback P_m has degree at least two and larger than the degrees of the others; with $P_j = z^{b^j}$ this covers linear Mahler equations. There the substitution acts on the variable of an entire series, not on the exponents of a scalar, and neither result is used for the other.

Camacho [2] shows that Hahn fields supplied with truncation or related normal-form predicates interpret monadic second-order arithmetic. The arithmetic consequence of knowing support is therefore prior work. The new proposed step is recovering that support structure from a *single named dilation*, with no named valuation or monomial group. We give a concrete Boolean-coefficient coding to make this consequence transparent. We do not answer Camacho’s monomial-definability question in the smaller language without the extra automorphism.

Multiplicative valued difference fields are also an established subject; see Pal [4]. Our coefficient action is the identity. We make no claim that these particular structures satisfy the residue difference closedness hypotheses used in relative quantifier-elimination results. The obstruction to $\sigma y - y = 1$ is already visible in the residue field.

The focused literature and repository review did not locate the complete orbit-resolvent/Koszul statement, the single-dilation recovery theorem, or the resulting unrestricted centralizer statement in the form proved here. These are *proposed additions*, not an assertion of exhaustive bibliographic priority. No named long-standing open problem is claimed resolved. In particular, nothing below determines the existence or image of nontrivial *exponential* automorphisms of **No**.

1.3 Repository snapshot and evidence boundary

The comparison is with the read-only snapshot

465a54b479a1ee842cbf7db1689a7d2f6bfe25e1.

The repository catalogue [6] lists a substantial collection of AI-assisted reports; its formalization ledger distinguishes paper proofs from checked declarations. The root README [5] documents strong Hahn summation, Neumann support control, coefficient maps, and infinitesimal exponential/logarithm infrastructure. Those are natural interfaces for formalizing this article.

The catalogue and the surcomplex-automorphism report’s guide were inspected, not every source line in the repository. The repository was not independently built. No theorem in this article is being represented as already checked by Lean. Appendix A records exactly what the delivered finite verification program does and does not check.

This subsection is the source’s, as of its pin. The repository statements it makes were checked again against the collection as it now stands; the result, with the one addition that matters here, is in Appendix D.2.

1.4 Notation and conventions in this report

This report prints the source manuscript with its notation adapted to the collection’s notation guide docs/NOTATION.md. One map was renamed and one formula symbol was renamed; no normalization changed.

1. *The exponent dilation is S_q .* For $q \in \mathbb{Q}_{>0}$ and divisible Γ (2-divisibility suffices for $q = 2$),

$$S_q\left(\sum_g a_g t^g\right) = \sum_g a_g t^{qg}.$$

The source wrote D_q , and in Section 8 and Appendix B abbreviated D_2 to D . The notation guide reserves D (with ∂) for a scalar derivation of the surreal field, and the collection’s surcomplex-automorphisms report writes this same map S_a (saut:eq:dilation, there for every positive surreal a) while using D_λ and D_χ for character twists $t^g \mapsto \chi(g)t^g$. Here S_q always carries its subscript, and the abbreviation D is not used: every D, D_2, D_q, D_r, D_s of the source is S with the same subscript (S_2 for the bare D). The only D left in the report is D_χ in Section 1.5, which denotes that report’s character twist. An unsubscripted S is a well-ordered subset of Γ (in the definition of K and in Lemma 3.1).

2. *The support formula is $\text{Supp}(m, x)$, “ m is a canonical monomial in the support of x ” (equation (8.5)).* The source wrote $S(m, x)$, which would now clash with the dilation. The support *set* is still $\text{supp}(x)$.

The other symbols are the source’s. Their meanings here are fixed as follows.

- σ is always a coefficient-fixing monomial automorphism $\sigma(t^g) = \chi(g)t^{\theta g}$ of the *scalar* field (2.1). It is not the argument dilation $f(z) \mapsto f(\lambda z)$ of formal functions studied in the collection’s holonomic-rigidity report, and not the symbolic shift or the small-divisor rate that the notation guide records under σ .
- $\theta \in \text{Aut}_{\text{ord}}(\Gamma)$ is the surcomplex-automorphisms report’s τ ; χ and ρ agree with its χ and ρ . In its notation $\sigma = M_{\text{id}, \theta, \chi}$ (saut:thm:monomial), and the maps (9.1) are its $M_{\rho, \theta} = M_{\rho, \theta, 1}$.
- $K = \mathbf{k}((t^\Gamma))$ is a *set-sized* full Hahn field over an arbitrary coefficient field $\mathbf{k}$ of characteristic zero; the guide’s $K_\Gamma = \mathbb{C}((t^\Gamma))$ is the case $\mathbf{k} = \mathbb{C}$. (The surcomplex-automorphisms report uses K for the class $\mathbf{No}(i)$.) The class $\mathbf{No}(i)$ is the guide’s $\mathbf{No}[i]$. In Section 13, and in Theorem 13.1(5) in particular, θ ranges over ordered additive class automorphisms of $(\mathbf{No}, +, <)$: there the exponent group is a proper class, departing from the guide’s set-sized Γ as the surcomplex-automorphisms report also does.
- $U_1 = 1 + \mathfrak{m}$ is the group of principal units. It is not that report’s group $\mathcal{U}$ of 1-automorphisms; the sets $\mathcal{U}_\pm(S)$ of Lemma 3.1 are orbit hulls of exponents. $M(m)$ is the monomial formula (8.3), not a map $M_{\rho, \tau}$. F is the fixed field of one σ (Corollary 4.2) or of a commuting family (5.1).
- *Resonance* means only the set $R_\lambda = \{g : \theta g = g, \chi(g) = \lambda\}$ and the restriction P_λ to it (4.1). It is unrelated to the multiplier resonances of the dynamics-and-normal-forms report. *Spectrum* is used once, in the algebraic sense stated after Corollary 4.2. A *centralizer* is always taken in a group of field automorphisms, not among germs.
- *Strong* means preserving Hahn summation (Definition in Section 2), not convergence of finite partial sums. *Dilation* always means an exponent dilation of the scalar field.

The tempting false readings, beside the true ones: S_2 is not squaring, since $S_2(\sum a_g t^g) = \sum a_g t^{2g}$ while $(\sum a_g t^g)^2 = \sum a_g a_h t^{g+h}$, and the two agree exactly on the canonical monomials (Lemma 8.2). S_q is not $f(z) \mapsto f(\lambda z)$, and it is not the surreal exponential, the omega-map, a derivation or an exponential-preserving automorphism (Section 13.3).

1.5 Relation to other reports in the collection

This subsection was written when the manuscript joined the collection; the source compared itself only with the root README, the catalogue and the surcomplex-automorphisms report's guide.

Surcomplex field automorphisms (surcomplex/surcomplex-field-automorphisms). The dilation S_q is that report's S_a at $a = q$ (saut:eq:dilation), and the maps (2.1) are its monomial automorphisms $M_{\rho,\tau,\chi}$ (saut:thm:monomial) with $\rho = \text{id}$ and $\tau = \theta$. The source credits that construction (Section 1) and uses none of that report's classification statements as premises. Four relations are precise.

1. *Consistency with the four-layer decomposition.* saut:thm:decomp factors every valued automorphism $\alpha \in G_v$ (those with $\alpha(O) = O$) of $\mathbf{No}(i)$ uniquely as $\alpha = u D_\chi M_{\rho,\tau}$. By Theorem 13.1(5), a class field automorphism of $\mathbf{No}(i)$ commuting with some S_q , $q \neq 1$, is of the form $M_{\rho,\theta}$. So it lies in G_v , and in that factorization $u = 1$ and $\chi = 1$. Theorem 9.1 is the set-sized analogue, and the computation after Corollary 9.2 is the same statement for the layer D_χ alone (D_χ commutes with S_2 only if $\chi = 1$). Both uniqueness statements describe the same maps; there is no conflict.
2. *One subgroup beyond G_v .* That report states, as non-claims, that its four-layer decomposition covers G_v only, not all field automorphisms, and that no complete classification of the automorphisms of $\mathbf{No}(i)$ is claimed (items 1 and 10 of saut:sec:nonclaims). The centralizer here is computed among *all* field automorphisms, with no assumption that they preserve the valuation ring or Hahn sums; being valued and strong is a conclusion. Thus one subgroup of the unrestricted group, the centralizer of S_q , is shown to lie in G_v and is identified there. That is the whole extension. It does not classify $\text{Aut}(\mathbf{No}(i))$, it says nothing about automorphisms that commute with no S_q , and both non-claims of that report otherwise stand.
3. *Complement to non-definability.* saut:prop:nondefinability shows that over every set of parameters there are pure-field automorphisms moving $\mathbf{No}$ and moving O , so neither is definable in the pure field with parameters. Theorem 8.1 and Theorem 13.1(4) show that after adding the single automorphism S_q the coefficient field, the monomials, every coefficient and the valuation ring become parameter-free definable. The expansion is a different structure, so the two results are complementary. On $\mathbf{No}(i)$ the real axis $\mathbf{No}$ is still not claimed definable, and the centralizer contains every automorphism of $\mathbb{C}$ (Section 13.3).
4. *Its questions.* None of the four questions of saut:sec:questions is answered here: the image of exponential automorphisms on the value group, omega-map compatibility, effective descriptions inside the leading-term kernel, and real-form and continuity classifications. The source says that it does not settle the exponential-automorphism image problems discussed in [1, 7] (Section 13.3). The only centralizer computed in that report is that of conjugation (saut:thm:axis), a different group; and that report never asserts that S_r and S_s are conjugate, so Theorem 11.3 contradicts nothing there.

Exponential automorphism rigidity (surreal/exponential-automorphism-rigidity). On $\mathbf{No}$, S_q is that report's canonical lift $\widehat{T}_q$ (prop:hahn-lift) of the value-group dilation $T_q(\gamma) = q\gamma$, and for rational $q \neq 1$ that report proves that T_q has no exponential lift (thm:dilation). With $\mathbf{k} = \mathbb{R}$, Theorem 13.1(5) says that the class field automorphisms of $\mathbf{No}$ commuting with S_q are exactly the canonical lifts $\widehat{T}$ of the ordered additive automorphisms T of $(\mathbf{No}, +, <)$. Theorems 11.2 and 11.3 are new statements about the same maps and contradict nothing there. Nothing here decides which of these lifts, or which other automorphisms, commute with the exponential; that report's Question 13.1 (q:image) is untouched.

Omnific-preserving automorphisms (surreal/omnific-preserving-automorphisms). *Related (batch 34)*. Part X of that report, from a manuscript that does not cite this one, re-derives the difference core of this report with trivial character. Its weighted Green operator (opa:us:thm:green) is the partial inverse of Theorem 4.1 with $\chi = 1$, for the monomial automorphism of any ordered additive automorphism of a set or class exponent group; the dictionary is opa:us:rem:dsup. Its proof works directly on the class **No** instead of localizing to a set-sized subgroup. For the automorphisms of its faithful actions of left-orderable groups on **No** and **No**(i), its exact multiplicative obstruction (opa:us:thm:multiplicative) and its relative multiplicative retraction (opa:us:thm:retraction) specialize Theorem 6.3; the retraction adds an explicit image of the inner difference operator. Its variable-coefficient criterion (opa:us:thm:firstorder) solves $\sigma x - ax = b$ for nonconstant a , beyond the constant-scalar resolvent here. Its central cocycle normal forms (opa:us:thm:cocycles) give the degree-one cohomology for every group with a central nonidentity element, where Theorem 7.1 computes every degree for $\mathbb{Z}^d$ acting by rational dilations. No statement of this report changes.

Holonomic rigidity (surcomplex/holonomic-rigidity-for-entire-hahn-functions). The dilations in that report's batch-22 material are argument dilations $f(z) \mapsto f(\lambda z)$ of formal functions, with λ in the field. The dilations here act on the exponents of scalars. The two share a word, not a theorem.

Birthday cutoffs and hereditary sets (foundations-and-computation/birthday-cutoffs-and-hereditary-sets), placed after the pin. There the ordered surreal field expanded by the birthday function interprets full second-order arithmetic (hset:prop:arithmetic), its complete theory is undecidable (hset:cor:undecidable), and a definable relation has the independence property (hset:prop:ip). Section 10 obtains the same kind of consequence, and in addition TP_2 and the strict order property, for a different expansion (a Hahn field with one exponent dilation) by a different mechanism (Boolean coefficient coding, after Camacho [2]). Neither result implies the other.

Unrelated uses of shared words. The resonances, homological operator and centralizers of surcomplex/dynamics-and-normal-forms concern local dynamics of germs, and the Koszul complexes of surcomplex/analytic-geometry and surcomplex/finite-deformations are complexes of polynomial ideals. No statement there is used or extended. No dsup: label has a Lean mapping in docs/FORMALIZATION.md.

2 Conventions, supports, and monomial automorphisms

Throughout the set-sized theory, Γ is an ordered abelian group, $\mathbf{k}$ is a field of characteristic zero, and

$$K = \mathbf{k}((t^\Gamma)) = \left\{ \sum_{g \in S} a_g t^g : S \subseteq \Gamma \text{ is well ordered, } a_g \in \mathbf{k} \right\}$$

is the *full* Hahn field. Zero coefficients can be omitted. We write x_g for the coefficient at g , $\text{supp}(x) = \{g : x_g \neq 0\}$, and

$$v(x) = \min \text{supp}(x) \quad (x \neq 0), \quad v(0) = \infty.$$

The coefficient field is embedded by $a \mapsto at^0$. The valuation ring, maximal ideal, and principal-unit group are

$$\mathcal{O} = \{x : v(x) \geq 0\}, \quad \mathfrak{m} = \{x : v(x) > 0\}, \quad U_1 = 1 + \mathfrak{m}.$$

The notation $[x]_0 = x_0$ always means the coefficient of t^0 , including for $x \notin \mathcal{O}$. It is not a ring homomorphism on all of K .

Definition 2.1 (Hahn summability). A set-indexed family $(x_i)_{i \in I}$ is *summable* if $\bigcup_i \text{supp}(x_i)$ is well ordered and, for each g , only finitely many i have $(x_i)_g \neq 0$. Its sum has coefficient $\sum_i (x_i)_g$. A $\mathbf{k}$ -linear map is *strong* if it takes summable families to summable families and commutes with their sums.

Restriction to any fixed subset of Γ is a strong linear map, since it only deletes support points. Strong summation permits infinite families whose finite subsums do not converge in the valuation topology. In particular, no argument below replaces a support proof by the assertion that an orbit “tends to infinity.”

Definition 2.2 (Coefficient-fixing monomial automorphism). Let $\theta \in \text{Aut}_{\text{ord}}(\Gamma)$ and let $\chi : \Gamma \rightarrow \mathbf{k}^\times$ be a group homomorphism. Put

$$\sigma \left(\sum_g a_g t^g \right) = \sum_g a_g \chi(g) t^{\theta g}. \quad (2.1)$$

Proposition 2.3. *The map (2.1) is a strong field automorphism fixing $\mathbf{k}$. Its valuation action is $v(\sigma x) = \theta(vx)$ for $x \neq 0$. Two such maps σ_i, σ_j commute if and only if*

$$\theta_i \theta_j = \theta_j \theta_i, \quad \chi_i(\theta_j g) \chi_j(g) = \chi_j(\theta_i g) \chi_i(g) \quad (g \in \Gamma). \quad (2.2)$$

Proof. An increasing bijection preserves well-ordered supports and point-finiteness. Additivity and strength follow coefficientwise. On monomials, $\chi(g+h) = \chi(g)\chi(h)$ and $\theta(g+h) = \theta g + \theta h$ give multiplicativity. Hahn convolution is legitimate because its coefficient sums are finite; thus the monomial calculation extends to arbitrary series. The inverse sends t^g to $\chi(\theta^{-1}g)^{-1} t^{\theta^{-1}g}$. Leading terms give the valuation formula. Computing both compositions on t^g proves (2.2), and strength extends the result to all series. $\square$

For $q \in \mathbb{Q}_{>0}$ on a divisible Γ , write S_q for the case $\theta g = qg$ and $\chi = 1$. For S_2 alone, 2-divisibility is sufficient. No ordinary absolute value on $\mathbf{k}$ is used in the Hahn theory.

3 The orbit-support lemma

The following elementary support fact is responsible for the unrestricted rank of the results. The forward direction of summation is chosen separately on the two movement chambers of θ .

Lemma 3.1 (Increasing orbit hull). *Let θ be an increasing automorphism of an ordered abelian group. If $S \subseteq \{g : \theta g > g\}$ is well ordered, then*

$$\mathcal{U}_+(S) = \{\theta^n s : s \in S, n \in \mathbb{N}\}$$

is well ordered, and every exponent has only finitely many representations $\theta^n s$ with $(n, s) \in \mathbb{N} \times S$. If $S \subseteq \{g : \theta g < g\}$ is well ordered, the same assertions hold for $\mathcal{U}_-(S) = \{\theta^{-n} s : s \in S, n \geq 1\}$.

Proof. Every sequence in a well-ordered set has an infinite nondecreasing subsequence. One proof uses tail minima: if a tail minimum occurs infinitely often, take those occurrences; otherwise pass beyond its last occurrence and continue with the larger minimum of the remaining tail. If this process does not stop, the chosen values strictly increase.

Suppose a strictly decreasing sequence in $\mathcal{U}_+(S)$ existed. Choose representations $\theta^{n_j} s_j$. By two successive subsequence selections, both n_j and s_j may be taken nondecreasing. For $j < k$,

$$\theta^{n_j} s_j \leq \theta^{n_j} s_k \leq \theta^{n_k} s_k,$$

because the forward orbit of s_k increases. This contradicts strict decrease. In ZFC, a linear order with no infinite strictly decreasing sequence is well ordered: a nonempty subset without a least element would recursively supply such a sequence.

If one exponent g had infinitely many representations, their n 's would be unbounded, since each n determines at most one s . Passing to strictly increasing n_j gives $s_j = \theta^{-n_j} g$. The point g is in the increasing chamber, which is θ -invariant, so these s_j strictly decrease. That contradicts the well-ordering of S .

For the second assertion apply the first to θ^{-1} on the decreasing chamber and omit the $n = 0$ terms. Every subset of a well-ordered set is well ordered. $\square$

Corollary 3.2 (Labeled version). *Suppose $(x_i)_{i \in I}$ is a summable family whose supports are all in the increasing chamber. Then $(\sigma^n x_i)_{(n,i) \in \mathbb{N} \times I}$ is summable. The corresponding inverse-iterate assertion holds on the decreasing chamber.*

Proof. Apply Lemma 3.1 to the union of the original supports. For a given output exponent there are finitely many pairs (n, s) ; at each s there are finitely many original labels i . Nonzero character factors and nonzero scalar multipliers do not affect this support argument. $\square$

Remark 3.3. The lemma is not an assertion of cofinal growth. For example, in $\mathbb{Q}^2$ with lexicographic order, the shear $\theta(a, b) = (a, b + a)$ takes $(1, 0)$ through $(1, n)$, a sequence bounded above by $(2, 0)$. Its orbit is summable in the Hahn sense, but finite subsums need not converge in the valuation topology.

4 An exact resolvent and its resonance obstruction

Fix σ as in (2.1) and $\lambda \in \mathbf{k}^\times$. Define

$$A_+ = \{g : \theta g > g\}, \quad A_- = \{g : \theta g < g\}, \quad A_\pm = \text{Fix}(\theta).$$

Write $f = f_+ + f_- + f_\pm$ by restricting its support to these three sets. The *resonance set* and *resonance projection* are

$$R_\lambda = \{g \in \Gamma : \theta g = g, \chi(g) = \lambda\}, \quad P_\lambda f = \sum_{g \in R_\lambda} f_g t^g. \quad (4.1)$$

Define

$$\begin{aligned} Q_\lambda f = & - \sum_{n \geq 0} \lambda^{-n-1} \sigma^n(f_+) + \sum_{n \geq 1} \lambda^{n-1} \sigma^{-n}(f_-) \\ & + \sum_{\substack{g \in A_\pm \\ \chi(g) \neq \lambda}} \frac{f_g}{\chi(g) - \lambda} t^g. \end{aligned} \quad (4.2)$$

All sums here are Hahn sums.

Theorem 4.1 (Strong partial inverse). *The map $Q_\lambda : K \rightarrow K$ is well defined, strong, and $\mathbf{k}$ -linear. It satisfies*

$$(\sigma - \lambda)Q_\lambda = Q_\lambda(\sigma - \lambda) = I - P_\lambda, \quad (4.3)$$

$$P_\lambda Q_\lambda = Q_\lambda P_\lambda = 0, \quad (4.4)$$

$$v(Q_\lambda f) \geq v(f). \quad (4.5)$$

Consequently,

$$\ker(\sigma - \lambda) = P_\lambda K, \quad \text{im}(\sigma - \lambda) = \ker P_\lambda. \quad (4.6)$$

The equation $(\sigma - \lambda)y = f$ is solvable exactly when $P_\lambda f = 0$. Its unique solution with $P_\lambda y = 0$ is $Q_\lambda f$.

Proof. Lemma 3.1 makes each of the first two families in (4.2) summable. The third is a restriction and coefficient rescaling of f . The finite union of the three supports is well ordered and the three families remain point-finite, so their sum exists. The labeled version proves strength and justifies interchanging any summable input family with the displayed sums.

On the increasing chamber, apply $\sigma - \lambda$ to the first sum:

$$-\sum_{n \geq 0} \lambda^{-n-1} \sigma^{n+1} f_+ + \sum_{n \geq 0} \lambda^{-n} \sigma^n f_+ = f_+.$$

The cancellation is coefficientwise between jointly summable families. There is no limiting remainder. On the decreasing chamber the calculation is

$$\sum_{n \geq 1} \lambda^{n-1} \sigma^{-(n-1)} f_- - \sum_{n \geq 1} \lambda^n \sigma^{-n} f_- = f_-.$$

On a fixed exponent, $\sigma - \lambda$ multiplies its coefficient by $\chi(g) - \lambda$. This proves the left-hand identity in (4.3). The chamber restrictions commute with σ , as do the fixed-exponent division operation and the iterate sums. Thus Q_λ commutes with σ , proving the right-hand identity. No term of (4.2) lies in R_λ , which gives (4.4).

Every exponent appearing in the first sum is at least its originating exponent, because it is a forward iterate in the increasing chamber. The same is true of inverse iterates in the decreasing chamber. Fixed exponents are unchanged. This proves (4.5), including the possibility of cancellation.

For completeness, the kernel also has a direct support proof. If $\sigma y = \lambda y$, then $\theta \text{supp}(y) = \text{supp}(y)$. A nonfixed exponent would therefore force its whole bi-infinite orbit into the support. That orbit contains an infinite strictly decreasing sequence. Hence the support is fixed pointwise by θ , and its coefficients require $\chi(g) = \lambda$. This gives the kernel in (4.6). The projection P_λ annihilates the image of $\sigma - \lambda$; conversely (4.3) produces every element of its kernel. Finally, two normalized solutions differ by a kernel element with zero resonance projection and therefore coincide. $\square$

Corollary 4.2 (Fixed fields and eigenspaces). *Let*

$$\Gamma_\sigma = \{g : \theta g = g, \chi(g) = 1\}, \quad F = \mathbf{k}((t^{\Gamma_\sigma})).$$

Then $K^\sigma = F$. If R_λ is nonempty and $g_\lambda \in R_\lambda$, then

$$R_\lambda = g_\lambda + \Gamma_\sigma, \quad \ker(\sigma - \lambda) = t^{g_\lambda} F.$$

For every $n \geq 1$, $\ker(\sigma - \lambda)^n = \ker(\sigma - \lambda)$.

Proof. The set Γ_σ is a subgroup, so its full Hahn series form a subfield. The fixed-field statement is Theorem 4.1 with $\lambda = 1$. Subtracting g_λ from another resonant exponent proves the coset formula, and translation of supports gives the eigenspace. The decomposition $K = P_\lambda K \oplus \ker P_\lambda$ is invariant under $\sigma - \lambda$, which is zero on its first summand and invertible on its second. Powers have the same kernel. $\square$

This is a statement about kernels and invertibility, not a claim that K is a direct sum of its eigenspaces. In particular, arbitrary series can have support on moving orbits. In the purely algebraic sense in which the spectrum of a linear map consists of scalars where it fails to be bijective, the spectrum of σ is $\chi(\text{Fix } \theta)$. Zero is not in this spectrum.

Corollary 4.3 (Split polynomial difference operators). *Suppose $p \in \mathbf{k}[X] \setminus \{0\}$ splits over $\mathbf{k}$. Put*

$$R_p = \{g : \theta g = g, p(\chi(g)) = 0\}, \quad P_p f = f \upharpoonright_{R_p}.$$

Then $\ker p(\sigma) = P_p K$ and $\text{im } p(\sigma) = \ker P_p$. Repeated roots of p do not enlarge the kernel. On $\ker P_p$, the inverse is strong.

Proof. The factor X is harmless since σ is a strong automorphism. On the moving chambers every nonzero scalar factor $\sigma - \lambda$ has the inverse supplied by (4.2); these inverses commute because they are inverses of commuting bijections on that invariant subspace. On the fixed chamber $p(\sigma)$ multiplies the coefficient at g by $p(\chi(g))$. Invert this scalar where nonzero and set the inverse to zero on R_p . The factors and all restrictions preserve summable families. $\square$

Remark 4.4 (Not a residue homomorphism). For $\lambda = 1$, P_1 is F -linear: multiplication by a fixed monomial translates supports by an element of Γ_σ , and does not change the resonance test; general F -linearity follows by Hahn convolution. Nevertheless, P_1 need not be multiplicative on K . For doubling and $g \neq 0$, both $P_1(t^g)$ and $P_1(t^{-g})$ vanish, whereas $P_1(t^g t^{-g}) = 1$.

5 Commuting equations and an explicit Koszul contraction

Let $\sigma_1, \dots, \sigma_d$ be a finite commuting family of coefficient-fixing monomial automorphisms. Write

$$\Delta_j = \sigma_j - I, \quad P_j = P_{j,1}, \quad Q_j = Q_{j,1}, \quad E_0 = I, \quad E_j = P_1 \cdots P_j, \quad P = E_d.$$

The subscript $(j, 1)$ refers to the construction in Theorem 4.1 for σ_j and $\lambda = 1$. Let

$$\Gamma_0 = \{g : \theta_j g = g, \chi_j(g) = 1 \text{ for every } j\}, \quad F = \mathbf{k}((t^{\Gamma_0})). \quad (5.1)$$

Thus P restricts a series to its joint-resonant exponents.

Lemma 5.1 (Commuting normalized inverses). *The maps P_i, Q_i, σ_i commute with P_j, Q_j, σ_j whenever the two displayed maps have different indices. In particular P commutes with every operator, and $F = K^{\sigma_1, \dots, \sigma_d}$.*

Proof. If g is fixed by θ_i , commutativity of the exponent maps makes $\theta_j g$ fixed by θ_i . The character identity (2.2) then gives $\chi_i(\theta_j g) = \chi_i(g)$. Thus the resonance subset for i is invariant under θ_j and its inverse. Restriction to this subset commutes with σ_j . The projections also commute with one another, being support restrictions.

Here is a useful uniqueness argument for the inverse. If a linear map A commutes with Δ_i and P_i , then $AQ_i f$ and $Q_i A f$ both have zero P_i -projection and both solve

$$\Delta_i z = (I - P_i) A f.$$

The normalized solution is unique, so $AQ_i = Q_i A$. Apply this first to $A = \sigma_j$ and $A = P_j$. It follows that Q_i commutes with Δ_j and P_j ; applying the same argument for Q_j then gives $Q_i Q_j = Q_j Q_i$. The common fixed field is obtained by intersecting the fixed-support conditions of Corollary 4.2. $\square$

Let $e_1, \dots, e_d$ be the standard basis of $\mathbf{k}^d$, and set

$$C^r = K \otimes_{\mathbf{k}} \bigwedge^r \mathbf{k}^d, \quad \mathbf{d} = \sum_{j=1}^d \Delta_j \varepsilon_j, \quad \varepsilon_j(\eta) = e_j \wedge \eta.$$

Here a cochain is only a finite tuple of Hahn series. The usual contraction ι_j on the exterior algebra is defined by

$$\iota_j(e_{i_1} \wedge \cdots \wedge e_{i_r}) = \begin{cases} (-1)^{s-1} e_{i_1} \wedge \cdots \widehat{e_{i_s}} \cdots \wedge e_{i_r}, & i_s = j, \\ 0, & j \notin \{i_1, \dots, i_r\}. \end{cases}$$

It satisfies $\varepsilon_i \iota_j + \iota_j \varepsilon_i = \delta_{ij} I$. Extend P to each cochain degree by applying it to every coefficient; call this extension Π .

Theorem 5.2 (Explicit deformation retraction). *The operator*

$$h = \sum_{j=1}^d Q_j E_{j-1} \iota_j : C^r \longrightarrow C^{r-1} \quad (5.2)$$

is strong on coefficient tuples and satisfies

$$dh + hd = I - \Pi, \quad h^2 = 0, \quad h\Pi = \Pi h = 0, \quad d\Pi = \Pi d = 0. \quad (5.3)$$

Consequently

$$H^r(\mathbb{Z}^d, K_{\text{add}}) \simeq F \otimes_{\mathbf{k}} \bigwedge^r \mathbf{k}^d \simeq F^{(d)}_{(r)} \quad (0 \leq r \leq d), \quad (5.4)$$

and the cohomology vanishes for $r > d$. The isomorphism sends a cocycle to its coefficientwise joint-resonant part.

Proof. All coefficient operators in (5.2) commute by Lemma 5.1. Therefore the exterior-algebra identities reduce the anticommutator to its diagonal terms:

$$\begin{aligned} dh + hd &= \sum_{j=1}^d \Delta_j Q_j E_{j-1} \\ &= \sum_{j=1}^d (I - P_j) E_{j-1} = \sum_{j=1}^d (E_{j-1} - E_j) = I - P. \end{aligned}$$

The contractions anticommute and square to zero, so $h^2 = 0$. Each term $Q_j E_{j-1}$ annihilates P because $Q_j P_j = 0$; commutativity gives both $h\Pi = 0$ and $\Pi h = 0$. Finally $\Delta_j P = P \Delta_j = 0$ for every j , which proves the remaining identities. Strength follows from strength of the finitely many factors and sums.

The target subcomplex $F \otimes \bigwedge^\bullet \mathbf{k}^d$ has zero differential. If $d\alpha = 0$, then $\alpha = dh\alpha + \Pi\alpha$, and a coboundary has zero projection. Thus the inclusion of that zero-differential complex induces exactly (5.4).

To identify this with group cohomology, regard K_{add} as a module over $\mathbb{Z}[z_1^{\pm 1}, \dots, z_d^{\pm 1}]$, with z_j acting by σ_j . The tensor product of the two-term free resolutions

$$0 \longrightarrow \mathbb{Z}[z_j^{\pm 1}] \xrightarrow{z_j - 1} \mathbb{Z}[z_j^{\pm 1}] \longrightarrow \mathbb{Z} \longrightarrow 0$$

is a free resolution of the trivial module for $\mathbb{Z}^d$. Exactness follows successively because, after imposing earlier relations, the next $z_j - 1$ remains a non-zero-divisor in a Laurent polynomial ring. Applying Hom into K_{add} produces the complex above, up to the standard consistent choice of exterior signs. It has length d , giving the assertion in larger degrees. $\square$

Corollary 5.3 (Simultaneous additive equations). *Let $f_1, \dots, f_d \in K$. The system*

$$\Delta_j y = f_j \quad (1 \leq j \leq d)$$

has a solution if and only if

$$\Delta_i f_j = \Delta_j f_i \quad \text{for all } i, j, \quad P f_j = 0 \quad \text{for all } j. \quad (5.5)$$

When these conditions hold, its unique solution with $P y = 0$ is

$$y = \sum_{j=1}^d Q_j E_{j-1} f_j. \quad (5.6)$$

All other solutions are obtained by adding an arbitrary element of F .

Proof. The first condition says that $\alpha = \sum_j f_j e_j$ is a cocycle. The second says $\Pi\alpha = 0$. Equation (5.3) gives $d(h\alpha) = \alpha$, and contraction of the basis vectors gives (5.6). Necessity follows from $d^2 = 0$ and $\Pi d = 0$. The normalization follows from $Ph = 0$, and uniqueness from the common fixed-field calculation. $\square$

Corollary 5.4 (Definable invariant projection). *In the language of fields with $\sigma_1, \dots, \sigma_d$, the graph of $P : K \rightarrow F$ is parameter-free definable by*

$$P(x) = c \iff \bigwedge_{j=1}^d \sigma_j c = c \wedge \exists y_1 \dots y_d \left(x - c = \sum_{j=1}^d \Delta_j y_j \right). \quad (5.7)$$

Proof. The operator identity $I - P = \sum_j \Delta_j Q_j E_{j-1}$ shows that the right-hand side holds for $c = Px$. Conversely P kills every $\Delta_j y_j$ and fixes $c \in F$, so applying P to the displayed equality gives $Px = c$. $\square$

Remark 5.5 (A cohomology calculation, not a false Leibniz rule). The operator Δ_j satisfies a twisted product rule, not the ordinary Leibniz rule. We use the Koszul complex as a complex of additive groups. No claim is made that its naive coefficientwise wedge multiplication makes d an ordinary derivation. This avoids confusing its additive calculation with a differential graded algebra structure.

6 Principal units and multiplicative difference equations

6.1 The formal logarithm is the appropriate logarithm

For $h \in \mathfrak{m}$, the formal series

$$\exp(h) = \sum_{n \geq 0} \frac{h^n}{n!}, \quad \log(1 + h) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} h^n \quad (6.1)$$

are strongly summable. The underlying support fact is the Neumann lemma: the additive monoid generated by a well-ordered subset of $\Gamma_{>0}$ is well ordered, with only finitely many contributions to a fixed exponent across all finite word lengths. This standard Hahn-series foundation, also part of the repository's documented summation layer [5], is used here in precisely its formal-support sense.

The formal power-series identities for exponential and logarithm can therefore be evaluated in $\mathfrak{m}$. Joint summability justifies substitution and regrouping. They give mutually inverse group isomorphisms

$$(\mathfrak{m}, +) \xrightleftharpoons[\log]{\exp} (U_1, \cdot).$$

A strong coefficient-fixing automorphism commutes with both maps, since it preserves the rational coefficients and every summable family in (6.1). This is an infinitesimal Hahn exponential. It is not a claim that a monomial automorphism commutes with the global surreal exponential.

Lemma 6.1 (Multiplicative coordinates). *Every $x \in K^\times$ has a unique expression*

$$x = c t^g \exp(h), \quad g = v(x), \quad c = \text{lc}(x) \in \mathbf{k}^\times, \quad h \in \mathfrak{m}. \quad (6.2)$$

Under σ_j , these coordinates transform as

$$(g, c, h) \mapsto (\theta_j g, c \chi_j(g), \sigma_j h). \quad (6.3)$$

Proof. Dividing x by its leading monomial ct^g leaves a principal unit; apply its unique logarithm. Multiplication adds the exponent and logarithm coordinates and multiplies the coefficient coordinate. Applying (2.1) and the naturality of (6.1) gives (6.3). $\square$

Put $\mathfrak{m}_F = F \cap \mathfrak{m}$. The projections P_j preserve $\mathfrak{m}$, and (4.5) shows that Q_j also preserves it. Thus the whole Koszul contraction restricts to infinitesimals.

Corollary 6.2 (Principal-unit cohomology). *For the commuting family of Section 5,*

$$H^r(\mathbb{Z}^d, \mathfrak{m}) \simeq \mathfrak{m}_F^{\binom{d}{r}}, \quad H^r(\mathbb{Z}^d, U_1) \simeq \mathfrak{m}_F^{\binom{d}{r}} \quad (0 \leq r \leq d),$$

where the second isomorphism uses $\log$. Both vanish above degree d .

6.2 One multiplicative equation

Theorem 6.3 (Exact multiplicative obstruction). *For one automorphism σ , write $f = c t^\gamma \exp(h)$ in multiplicative coordinates and let $P = P_1$, $Q = Q_1$. The equation*

$$\frac{\sigma y}{y} = f, \quad y \in K^\times, \quad (6.4)$$

is solvable if and only if there exists $g \in \Gamma$ with

$$(\theta - I)g = \gamma, \quad \chi(g) = c, \quad Ph = 0. \quad (6.5)$$

For each such g , $y = t^g \exp(Qh)$ is a solution. The full solution set, when nonempty, is a torsor under $(K^\sigma)^\times$.

Proof. Write $y = d t^g \exp(u)$. Formula (6.3) gives

$$\frac{\sigma y}{y} = \chi(g) t^{(\theta - I)g} \exp((\sigma - I)u). \quad (6.6)$$

Uniqueness of multiplicative coordinates proves necessity of the first two conditions and reduces the third to $\Delta u = h$. Theorem 4.1 solves that equation exactly when $Ph = 0$, with $u = Qh \in \mathfrak{m}$. The ratio of any two solutions is fixed by σ , and multiplication by a fixed nonzero element preserves a solution. $\square$

For a quotient description, use componentwise addition/multiplication on $\Gamma \times \mathbf{k}^\times$. The map

$$g \mapsto ((\theta - I)g, \chi(g))$$

is a group homomorphism. The preceding theorem yields

$$K^\times / \{\sigma y / y : y \in K^\times\} \simeq \frac{\Gamma \times \mathbf{k}^\times}{\{((\theta - I)g, \chi(g)) : g \in \Gamma\}} \times \mathfrak{m}_{K^\sigma}. \quad (6.7)$$

The isomorphism sends the class of $ct^\gamma \exp(h)$ to the class of (γ, c) and to Ph . Its kernel is exactly the displayed subgroup. In general the first quotient cannot be replaced by independent “valuation” and “leading coefficient” quotients: the same g must solve both equations in (6.5).

6.3 A commuting multiplicative system

A tuple $(f_1, \dots, f_d)$ in $(K^\times)^d$ is multiplicatively compatible if

$$\frac{\sigma_i(f_j)}{f_j} = \frac{\sigma_j(f_i)}{f_i} \quad \text{for all } i, j. \quad (6.8)$$

This is precisely the consistency condition forced by $\sigma_j y / y = f_j$.

Theorem 6.4 (Simultaneous multiplicative equations). *Write $f_j = c_j t^{\gamma_j} \exp(h_j)$. Suppose (6.8) holds. Then there exists $y \in K^\times$ with $\sigma_j y / y = f_j$ for every j if and only if*

$$\begin{aligned} \text{there is one } g \in \Gamma \text{ such that } (\theta_j - I)g &= \gamma_j, \quad \chi_j(g) = c_j \quad (1 \leq j \leq d), \\ Ph_j &= 0 \quad (1 \leq j \leq d). \end{aligned} \quad (6.9)$$

When these conditions hold, a solution is

$$y = t^g \exp \left(\sum_{j=1}^d Q_j E_{j-1} h_j \right), \quad (6.10)$$

and all solutions differ by an element of $F^\times$.

Proof. Applying uniqueness of multiplicative coordinates to (6.8) gives exactly

$$\begin{aligned} (\theta_i - I)\gamma_j &= (\theta_j - I)\gamma_i, \\ \chi_i(\gamma_j) &= \chi_j(\gamma_i), \\ \Delta_i h_j &= \Delta_j h_i. \end{aligned}$$

The constants c_j cancel from compatibility, but not from the search for a common solution. Formula (6.6), for all j , shows that the exponent and coefficient coordinates require a single g as in (6.9). The remaining additive system in $\mathfrak{m}$ is solved by Corollary 5.3 and its infinitesimal restriction. This gives (6.10). Ratios of solutions are fixed by every σ_j . $\square$

For clarity about the residual cohomology, let $M = \Gamma \times \mathbf{k}^\times$ with action $(g, c) \mapsto (\theta_j g, \chi_j(g)c)$. Lemma 6.1 is an equivariant isomorphism of abelian groups $K^\times \simeq M \times \mathfrak{m}$. Hence

$$H^r(\mathbb{Z}^d, K^\times) \simeq H^r(\mathbb{Z}^d, M) \times \mathfrak{m}_F^{\binom{d}{r}}. \quad (6.11)$$

This separates the fully computed principal-unit obstruction from the ordinary finite-coordinate monomial-module problem. We do not claim that $H^r(\mathbb{Z}^d, M)$ has a universal smaller formula for arbitrary shears and characters.

7 Rational dilations: only scalar obstructions survive

Assume in this section that Γ is divisible. Let $q_1, \dots, q_d \in \mathbb{Q}_{>0}$, with at least one $q_j \neq 1$, and use $\sigma_j = S_{q_j}$. Since a nonzero rational scalar acts injectively on an ordered divisible group, the joint fixed exponent subgroup is $\{0\}$. Therefore

$$F = \mathbf{k}, \quad P(x) = [x]_0, \quad \mathfrak{m}_F = 0. \quad (7.1)$$

Theorem 7.1 (Dilation equations and cohomology). *For this family,*

$$H^r(\mathbb{Z}^d, K_{\text{add}}) \simeq \mathbf{k}^{\binom{d}{r}}, \quad (7.2)$$

$$H^r(\mathbb{Z}^d, K^\times) \simeq (\mathbf{k}^\times)^{\binom{d}{r}} \quad (0 \leq r \leq d). \quad (7.3)$$

Both vanish for $r > d$. A compatible additive system is solvable exactly when every right-hand side has constant coefficient zero. A compatible multiplicative system is solvable exactly when every right-hand side has leading coefficient one.

Proof. The additive formula and its solvability criterion follow from (7.1). In multiplicative coordinates, the monomial module is now the direct product of Γ acted on by rational multiplication and $\mathbf{k}^\times$ acted on trivially. On Γ , at least one $q_j - 1$ is an automorphism. Its inverse times the corresponding exterior contraction contracts the entire Koszul complex. Thus all cohomology of this Γ -module, including invariants, is zero. The infinitesimal module is also acyclic by Corollary 6.2. The trivial module $\mathbf{k}^\times$ has zero Koszul differential, giving (7.3).

For an explicit check of the multiplicative criterion, choose j_0 with $q_{j_0} \neq 1$ and set $g = \gamma_{j_0}/(q_{j_0} - 1)$. Compatibility gives

$$(q_{j_0} - 1)\gamma_j = (q_j - 1)\gamma_{j_0},$$

so $\gamma_j = (q_j - 1)g$ for every j . The character equations are exactly $c_j = 1$, and the logarithmic obstruction vanishes because $\mathfrak{m}_F = 0$. This proves both necessity and sufficiency without an appeal to the abstract quotient description. $\square$

For one $q \neq 1$, these statements read

$$S_q y - y = f \iff [f]_0 = 0 \quad \text{for existence of } y \in K, \quad (7.4)$$

$$S_q y / y = f \iff \text{lc}(f) = 1 \quad \text{for existence of } y \in K^\times. \quad (7.5)$$

The solutions are unique modulo addition of $\mathbf{k}$ or multiplication by $\mathbf{k}^\times$, respectively. These are existence equivalences, not claims that either equation has a solution for every right-hand side.

Example 7.2 (Two opposite support directions). For S_2 one has

$$(S_2 - I) \left(- \sum_{n \geq 0} t^{2^n} \right) = t, \quad (S_2 - I) \left(\sum_{n \geq 1} t^{-2^{-n}} \right) = t^{-1}.$$

The second solution has negative exponents increasing to zero. It is a perfectly valid Hahn series, but it is not a Laurent series with a fixed bound on exponent denominators. There is no solution to $(S_2 - I)y = 1$.

Example 7.3 (A multiplicative Mahler product). The family defining

$$y = \prod_{n \geq 0} (1 - t^{2^n})^{-1}$$

is interpreted either coefficientwise or by its strongly summable logarithm. It satisfies $S_2 y / y = 1 - t$: doubling drops the factor with $n = 0$. On the other hand, $S_2 y / y = 2$ has no nonzero solution. The obstruction is its leading coefficient, not its valuation. These particular rank-one identities are illustrative Mahler calculations, not separate claims of novelty.

8 Doubling defines the entire coefficient structure

For this section assume only that $\Gamma \neq 0$ is 2-divisible. (The source abbreviated S_2 to D in this section; the report writes S_2 throughout, see Section 1.4.) We work in the one-sorted language

$$L_{S_2} = \{0, 1, +, -, \cdot, S_2\}.$$

Division by a nonzero element is a definable operation in this language. Every auxiliary symbol introduced below abbreviates an explicit first-order formula. Neither the coefficient field nor the value group is named in L_{S_2} .

Theorem 8.1 (Recovery from a single dilation). *In (K, S_2) the following data are parameter-free definable: the embedded coefficient field, the constant-coefficient map, the canonical monomial group t^Γ , the coefficient function on $K \times t^\Gamma$, the support relation, the order of the monomial exponents, the valuation ring and its maximal ideal, the leading monomial and coefficient, and truncation at any monomial exponent.*

The proof is constructive. We give the definitions and check each one.

8.1 Constants and the constant-coefficient map

By Corollary 4.2, since $2g = g$ only for $g = 0$,

$$C(c) : S_2c = c \quad (8.1)$$

defines precisely $\mathbf{k}$. Theorem 4.1 gives

$$B(x, c) : S_2c = c \wedge \exists y (x - c = S_2y - y). \quad (8.2)$$

For every x there is exactly one c satisfying $B(x, c)$, namely $[x]_0$. Write $B(x)$ for this definable function. Only the inverse of doubling on Γ is needed in that application of the resolvent theorem.

8.2 The equation that recognizes monomials

Lemma 8.2 (Monomial recognition). *For $x \in K$,*

$$x \in t^\Gamma \iff x \neq 0 \wedge S_2x = x^2. \quad (8.3)$$

Proof. Every t^g satisfies $S_2(t^g) = t^{2g} = (t^g)^2$. Conversely, let $x \neq 0$ satisfy the equation. Write $x = ct^g(1 + u)$, where $c \in \mathbf{k}^\times$ and $u \in \mathfrak{m}$. Comparison of leading coefficients gives $c = c^2$, so $c = 1$. The remaining equation is $S_2(1 + u) = (1 + u)^2$. If $u \neq 0$, put $\varepsilon = v(u) > 0$. The left-hand side minus 1 has valuation 2ε , whereas $2u + u^2$ has valuation ε , since the characteristic is zero and $v(u^2) = 2\varepsilon > \varepsilon$. This is impossible. Thus $u = 0$ and $x = t^g$. $\square$

Let $M(m)$ denote the formula on the right of (8.3). An important feature is that it selects coefficient *one*, not arbitrary nonzero scalar multiples of monomials. It is also entirely algebraic; no positivity condition on x occurs.

8.3 All coefficients and the support relation

For $M(m)$ define

$$\text{coef}(x, m) = B(x/m). \quad (8.4)$$

If $m = t^g$, this equals x_g . Hence

$$\text{Supp}(m, x) : M(m) \wedge \text{coef}(x, m) \neq 0 \quad (8.5)$$

is the exact support relation expressed on monomials. In particular,

$$x = y \iff \forall m (M(m) \Rightarrow \text{coef}(x, m) = \text{coef}(y, m)). \quad (8.6)$$

This step recovers the full series, not merely its leading term.

8.4 Recovering the orientation of the valuation

Define

$$J(m) : M(m) \wedge m \neq 1 \wedge B((1-m)^{-1}) = 1. \quad (8.7)$$

For $m = t^g$ with $g > 0$, the expansion $(1-m)^{-1} = \sum_{n \geq 0} m^n$ has constant coefficient one. For $g < 0$, the correct expansion is

$$(1-m)^{-1} = - \sum_{n \geq 1} m^{-n},$$

whose constant coefficient is zero. Both supports are well ordered in their indicated directions. Thus

$$J(t^g) \iff g > 0. \quad (8.8)$$

Consequently, for monomials m, n ,

$$v(m) < v(n) \iff J(n/m). \quad (8.9)$$

Multiplication of monomials corresponds to addition in Γ , so this defines the entire ordered value group on the definable set M . For real coefficients its order is the reverse of the usual order of positive monomials. For complex coefficients no field order is required.

8.5 The valuation ring, leading term, and truncation

The following formulas now define O and $\mathfrak{m}$:

$$x \in O \iff \forall m (\text{Supp}(m, x) \Rightarrow (m = 1 \vee J(m))), \quad (8.10)$$

$$x \in \mathfrak{m} \iff \forall m (\text{Supp}(m, x) \Rightarrow J(m)). \quad (8.11)$$

They include zero by vacuous quantification, as they should. For $x \neq 0$, the leading monomial is the unique ℓ such that

$$\text{Supp}(\ell, x) \wedge \forall m (\text{Supp}(m, x) \Rightarrow (m = \ell \vee J(m/\ell))). \quad (8.12)$$

Its coefficient is $\text{coef}(x, \ell)$. Existence follows from the well-ordering of the support.

For $M(m)$, the truncation $T(x, m)$ removes all exponents at least $v(m)$:

$$T(x, t^g) = \sum_{h < g} x_h t^h.$$

Its graph is given by

$$y = T(x, m) \iff M(m) \wedge \forall n (M(n) \Rightarrow [(J(m/n) \Rightarrow \text{coef}(y, n) = \text{coef}(x, n)) \wedge (\neg J(m/n) \Rightarrow \text{coef}(y, n) = 0)]). \quad (8.13)$$

The retained support is a subset of a well-ordered set, so the required y exists. Equation (8.6) gives uniqueness. This completes the proof of Theorem 8.1.

Remark 8.3 (The role of fullness). The proof uses that the orbit sums solving $S_2 y - y = f - [f]_0$ belong to the field. An arbitrary subfield of $\mathbf{k}((t^\Gamma))$ that is stable under S_2 need not have this property. Likewise, all Boolean subseries used below must exist. Thus replacing the full Hahn field by a rational function field, a finite-support ring, or an arbitrary union of Laurent fields requires new hypotheses. Definability in the full field does not supply an algorithm for arbitrary presentations of its elements.

9 The full centralizer of doubling

The defining formulas have a strong algebraic consequence. A field automorphism commuting with doubling must preserve all the recovered normal-form data, even when nothing of that kind is assumed beforehand.

Theorem 9.1 (Unrestricted centralizer). *Let $\Gamma \neq 0$ be 2-divisible. Every field automorphism $\alpha \in \text{Aut}(K)$ satisfying $\alpha S_2 = S_2 \alpha$ has a unique form*

$$\alpha \left(\sum_g a_g t^g \right) = \sum_g \rho(a_g) t^{\theta g}, \quad \rho \in \text{Aut}(\mathbf{k}), \quad \theta \in \text{Aut}_{\text{ord}}(\Gamma). \quad (9.1)$$

Conversely, every map (9.1) commutes with S_2 . Therefore

$$\text{Cen}_{\text{Aut}(K)}(S_2) \simeq \text{Aut}(\mathbf{k}) \times \text{Aut}_{\text{ord}}(\Gamma). \quad (9.2)$$

Every member of this centralizer is strong and preserves the valuation ring and all truncations.

Proof. Since α is an automorphism of the L_{S_2} -structure, it preserves every parameter-free definable set and function in Theorem 8.1. In particular its restriction to the coefficient field is an automorphism ρ of $\mathbf{k}$. It permutes the canonical monomials, so write $\alpha(t^g) = t^{\theta g}$. Multiplicativity makes θ additive; bijectivity makes it a group automorphism. Preservation of J gives $g > 0 \iff \theta g > 0$, hence θ is increasing.

For every $x \in K$ and $g \in \Gamma$, preservation of the coefficient function gives

$$\text{coef}(\alpha x, t^{\theta g}) = \text{coef}(\alpha x, \alpha(t^g)) = \alpha(\text{coef}(x, t^g)) = \rho(x_g).$$

The map θ is onto, so this computes every coefficient of αx . It proves (9.1) by uniqueness of Hahn expansions. There is no prior assumption that α commutes with infinite sums. The formula itself now proves that it does: θ preserves well-ordered supports, while ρ preserves finite sums and nonzero coefficients.

Conversely, (9.1) defines a strong field automorphism. Additivity of θ gives $\theta(2g) = 2\theta(g)$, so it commutes with S_2 . The two choices are unique on coefficients and monomials. Their composition is componentwise, since coefficient and exponent operations commute, which gives the direct product in (9.2). $\square$

Corollary 9.2 (No hidden centralizing motions). *A field automorphism commuting with S_2 and fixing every coefficient and canonical monomial is the identity. In particular a centralizing automorphism which fixes every leading term is the identity.*

Proof. For the first assertion use (9.1) with $\rho = \text{id}$ and $\theta = \text{id}$. For the second, the formula implies that fixing all leading terms fixes the coefficient and exponent components, so the first assertion applies. $\square$

A character twist $t^g \mapsto \chi(g)t^g$ illustrates the force of the condition: it commutes with doubling only if $\chi(2g) = \chi(g)$, but the character law says $\chi(2g) = \chi(g)^2$. Thus $\chi(g) = 1$ for every g . The theorem also excludes centralizing automorphisms which are not monomial twists and might initially have been nonvalued or nonstrong.

When $\mathbf{k} = \mathbb{R}$, $\text{Aut}(\mathbf{k})$ is trivial. Indeed an automorphism of $\mathbb{R}$ preserves positivity because positive elements are exactly nonzero squares, fixes $\mathbb{Q}$, and therefore fixes each real number by its rational cut. When $\mathbf{k} = \mathbb{C}$, arbitrary field automorphisms of $\mathbb{C}$ remain; only those equal to the identity or complex conjugation preserve $\mathbb{R}$ setwise, by the preceding observation and $\mathbb{C} = \mathbb{R}(i)$.

10 Arithmetic interpreted by one automorphism

This section identifies the logical cost of recovering normal-form information. The implication from support to arithmetic is related to Camacho's theorem [2]; the formulas below show directly how it is implemented in the smaller, single-dilation signature. The elementary coding is included so the dependence on full Hahn supports is explicit.

10.1 A uniform Boolean-series interpretation

Choose a parameter m satisfying $J(m)$; such a monomial exists since $\Gamma \neq 0$. Set $b_m = (1 - m)^{-1}$. Its support is exactly $\{m^n : n \in \mathbb{N}\}$. Define

$$N_m = \{u : M(u) \text{ and } \text{coef}(b_m, u) \neq 0\}.$$

Field multiplication on N_m represents addition of natural numbers: $m^r m^s = m^{r+s}$. The represented zero and one are 1 and m . Define the Boolean-series set

$$\mathcal{B}_m = \{x \in K : \forall u [M(u) \Rightarrow (\text{coef}(x, u) = 0 \vee \text{coef}(x, u) = 1)] \wedge \forall u [\text{Supp}(u, x) \Rightarrow u \in N_m]\}. \quad (10.1)$$

Each subset $A \subseteq \mathbb{N}$ has exactly one representative $x_A = \sum_{n \in A} m^n$ in $\mathcal{B}_m$. Conversely every member of $\mathcal{B}_m$ has this form. Membership is the definable relation $\text{coef}(x_A, m^n) = 1$. Thus the two definable sorts $N_m, \mathcal{B}_m$, with these operations and relation, are exactly $(\mathbb{N}, \mathcal{P}(\mathbb{N}); +, \in)$. No quotient by equal supports is needed because Boolean coefficients make the coding unique.

Theorem 10.1 (Undecidability in the unexpanded signature). *For every nonzero 2-divisible Γ and characteristic-zero field $\mathbf{k}$, the complete first-order theory of $(\mathbf{k}((t^\Gamma)), S_2)$ is undecidable. It uniformly interprets full monadic second-order arithmetic with one suitable monomial parameter. The undecidability assertion does not require adding a constant symbol for that parameter.*

Proof. The preceding definitions give an effective translation of every sentence φ of $(\mathbb{N}, \mathcal{P}(\mathbb{N}); +, \in)$ to a formula φ^m with the parameter m . Every m satisfying J yields the same standard interpreted structure. Therefore

$$(\mathbb{N}, \mathcal{P}(\mathbb{N}); +, \in) \models \varphi \iff (K, S_2) \models \exists m (J(m) \wedge \varphi^m). \quad (10.2)$$

For completeness, multiplication on $\mathbb{N}$ is definable in the interpreted structure. The relation $a \mid b$ says that every subset containing zero and closed under addition by a also contains b . This is exactly membership in $a\mathbb{N}$, because that set is the least such subset. For $a > 0$, the least positive common multiple of a and $a + 1$ is $a(a + 1)$: consecutive positive integers are coprime. Leastness is expressible with the order defined from addition. For $a = 0$ define this product to be zero. Its difference from a defines a^2 . Finally

$$(a + b)^2 = a^2 + b^2 + 2ab$$

defines ab uniquely using addition and squares. Hence ordinary true arithmetic is interpreted and its undecidability rules out a decision algorithm for the theory of (K, S_2) via (10.2). The explicit existential quantification is important: interpretation with an arbitrary named parameter alone would not justify this particular reduct claim. $\square$

The interpreted second sort is genuinely all of $\mathcal{P}(\mathbb{N})$, not just computable sets or eventually periodic sets. This is why full Hahn series matter. The theorem concerns the complete theory of the specified structure; it does not assert a failure of decidability for every smaller language associated with a valued field.

10.2 A direct coefficient witness to TP_2

The following argument is independent of the arithmetic reduction and uses arbitrary coefficients rather than only zero and one.

Theorem 10.2 (Coefficient independence). *The formula*

$$\varphi(x; u, c) : M(u) \wedge C(c) \wedge \text{coef}(x, u) = c$$

has the tree property of the second kind in the complete theory of (K, S_2) . The support relation also has the independence property and the strict order property.

Proof. Fix m with $J(m)$ and use parameters $(u_i, c_j) = (m^i, j)$ for $i, j \in \mathbb{N}$, where j is its characteristic-zero image in $\mathbf{k}$. For each fixed row i , the formulas $\varphi(x; m^i, j)$ for different j are pairwise inconsistent: a coefficient has only one value. For every function $\eta : \mathbb{N} \rightarrow \mathbb{N}$, the Hahn series

$$x_\eta = \sum_{i \geq 0} \eta(i) m^i$$

exists and satisfies all formulas $\varphi(x; m^i, \eta(i))$. Its support is a subset of the well-ordered set $\{i v(m) : i \in \mathbb{N}\}$, regardless of the sizes of its coefficients. This array is exactly a TP_2 witness, with two-inconsistent rows and every choice of one entry in each row realized.

The Boolean series x_A above shatter the infinite set $\{m^i : i \in \mathbb{N}\}$ under the support relation, giving the independence property. Finally the supports of $1 + m + \dots + m^n$ form a strictly increasing chain of definable sets of monomials. This witnesses the strict order property. $\square$

Remark 10.3 (No contradiction with tame valued-field reducts). Adding S_2 is a genuine expansion. The formulas use S_2 to recover coefficient extraction throughout the field, including its purely infinite part. The resulting interpreted arithmetic does not follow merely from naming a valuation ring. Likewise, this coefficient-fixing structure is not being identified with the existentially closed residue difference structures occurring in other theories of valued difference fields [4].

11 Every nonidentity rational dilation recovers the same data

Assume again that $\Gamma \neq 0$ is divisible. For $q = a/b \in \mathbb{Q}_{>0} \setminus \{1\}$, with positive integers a, b , let $n = |a - b| > 0$ and put

$$E_{a,b}(x) : x \neq 0 \wedge S_q(x)^b = x^a.$$

The next result both extends the recovery theorem and distinguishes interdefinability from conjugacy.

Lemma 11.1 (Power-equation recognition). *In (K, S_q) ,*

$$\{x : E_{a,b}(x)\} = \mu_n(\mathbf{k}) t^\Gamma, \quad t^\Gamma = \{y^n : E_{a,b}(y)\},$$

where $\mu_n(\mathbf{k}) = \{c \in \mathbf{k}^\times : c^n = 1\}$.

Proof. Write $x = ct^g \exp(h)$. The exponent equation is automatic since $bq = a$. The coefficient equation is $c^b = c^a$, equivalently $c^n = 1$. The principal-unit equation is $bS_q h = ah$, or $S_q h = qh$. The resonance set for the scalar $\lambda = q$ is empty: the only fixed exponent is zero and its character value is $1 \neq q$. Therefore Theorem 4.1 implies $h = 0$. This proves the first equality. The n th powers of these elements are precisely $t^{n\Gamma} = t^\Gamma$, because Γ is divisible and every $c \in \mu_n(\mathbf{k})$ has $c^n = 1$. $\square$

Theorem 11.2 (Interdefinable dilations). *For each nonidentity $q \in \mathbb{Q}_{>0}$, the structure (K, S_q) parameter-free defines all the normal-form data of Theorem 8.1 and every rational dilation S_r . In particular,*

$$\text{Cen}_{\text{Aut}(K)}(S_q) = \{\text{untwisted coefficient-ordered-exponent lifts}\} \simeq \text{Aut}(\mathbf{k}) \times \text{Aut}_{\text{ord}}(\Gamma).$$

These centralizers are the same subgroup of $\text{Aut}(K)$ for all $q \neq 1$. The arithmetic and TP_2 conclusions hold for every such S_q .

Proof. The fixed field of S_q is $\mathbf{k}$, and $x - c \in \text{im}(S_q - I)$ defines $c = [x]_0$, by (7.4). Lemma 11.1 defines the canonical monomial set. Once these two pieces are known, every subsequent formula in Section 8 is unchanged.

For a positive rational $r = c/d$, the monomial power m^r is the unique monomial z satisfying $z^d = m^c$; existence uses divisibility of Γ and uniqueness uses its torsion-freeness. The graph of S_r is then defined by

$$y = S_r x \iff \forall m (M(m) \Rightarrow \text{coef}(y, m^r) = \text{coef}(x, m)).$$

Every exponent is an r -multiple, so this determines all coefficients. Existence follows from the increasing exponent automorphism $g \mapsto rg$. In particular S_2 is definable from S_q , and conversely S_q is definable from S_2 . The centralizer statement follows either from these mutual definitions or by repeating the proof of Theorem 9.1. The logical consequences transfer through the explicit definitions. $\square$

Theorem 11.3 (Distinct rational dilations are not conjugate). *If $r, s \in \mathbb{Q}_{>0}$ are distinct, then S_r and S_s are not conjugate by any field automorphism of K . Moreover,*

$$\text{Nor}_{\text{Aut}(K)}(\langle S_2 \rangle) = \text{Cen}_{\text{Aut}(K)}(S_2).$$

Proof. The identity dilation is not conjugate to a nonidentity map, so assume $r \neq 1$. Write $r = a/b$ and $n = |a - b|$. The equation

$$S_r(x)^b = x^a, \quad x \neq 0,$$

has infinitely many solutions, since every t^s is a solution and $\Gamma \neq 0$ is infinite. For S_s in place of S_r , the valuation comparison gives $(bs - a)v(x) = 0$, so $v(x) = 0$ because $s \neq r$. Writing $x = c \exp(h)$ reduces the equation to $c^n = 1$ and $S_s h = rh$. There is no nonzero solution of the latter scalar eigen-equation: if $s \neq 1$, its only possible fixed exponent is zero, with eigenvalue $1 \neq r$; if $s = 1$, it is simply $(1 - r)h = 0$. Thus there are at most n nonzero solutions for S_s . Conjugation would carry the first solution set bijectively to the second, a contradiction. Equivalently, a single first-order sentence requiring more than n distinct solutions distinguishes the two pointed fields.

The group $\langle S_2 \rangle$ is infinite cyclic. An automorphism normalizing it must send its generator to S_2 or $S_2^{-1} = S_{1/2}$. The second alternative is excluded by nonconjugacy, leaving exactly the centralizer. $\square$

Interdefinability and nonconjugacy are compatible. Interdefinability recovers each particular dilation by a formula from another; conjugacy would have to carry the distinguished operator itself to the other one. The power equations distinguish the latter requirement.

12 Arbitrary-rank examples and sharp distinctions

12.1 A shear with a large fixed Hahn field

Take $\Gamma = \mathbb{Q}^2$ in lexicographic order and

$$\theta(a, b) = (a, b + a), \quad \chi = 1.$$

This is an increasing group automorphism, with inverse $(a, b) \mapsto (a, b - a)$. Its increasing and decreasing chambers are $a > 0$ and $a < 0$, respectively. The fixed exponent subgroup is $\{0\} \times \mathbb{Q}$, so

$$K^\sigma = \mathbf{k}((t^{\{0\} \times \mathbb{Q}})).$$

The entire series supported on that subgroup is the additive obstruction, not just the scalar constant coefficient. For example,

$$(\sigma - I) \left(- \sum_{n \geq 0} t^{(1,n)} \right) = t^{(1,0)}, \quad - \sum_{n \geq 0} t^{(1,n)} = - \frac{t^{(1,0)}}{1 - t^{(0,1)}}. \quad (12.1)$$

The inverse is exact although the exponents $(1, n)$ never exceed $(2, 0)$. The errors in finite partial sums therefore do not eventually have valuation greater than $(2, 0)$. Equation (12.1) is a concrete reason to retain strong support arguments rather than impose a rank-one analytic convergence intuition.

The additive and multiplicative obstructions are genuinely different. A nonzero fixed monomial $t^{(0,b)}$ cannot be in the additive image of $\sigma - I$. Nevertheless it is a multiplicative coboundary:

$$\frac{\sigma(t^{(b,0)})}{t^{(b,0)}} = t^{(0,b)}.$$

Being fixed does not itself prevent multiplicative exactness.

12.2 A phase character and an integral-exponent obstruction

Take $K = \mathbb{C}((t^{\mathbb{Q}}))$, $\theta = \text{id}$, and $\chi(g) = e^{2\pi i g}$. Then

$$\sigma \left(\sum_g a_g t^g \right) = \sum_g a_g e^{2\pi i g} t^g, \quad K^\sigma = \mathbb{C}((t^{\mathbb{Z}})).$$

Here every exponent is fixed by θ , so solving $(\sigma - I)y = f$ means deleting no moving orbits at all. It is possible exactly when every integral-exponent coefficient of f vanishes; the other coefficients are divided by $e^{2\pi i g} - 1$.

These denominators may be arbitrarily small as ordinary complex numbers. They do not obstruct strong Hahn summability: no support points are introduced and no infinite coefficientwise sums occur. Thus analytic small-denominator estimates and Hahn support obstructions are different issues.

For a multiplicative right-hand side $f = ct^\gamma \exp(h)$, the criterion is

$$\gamma = 0, \quad c \text{ is a root of unity}, \quad h \downarrow_{\mathbb{Z}} = 0.$$

Indeed $(\theta - I)g = 0$, and $\chi(\mathbb{Q})$ is exactly the group of complex roots of unity. In the last condition $h \in \mathfrak{m}$, so only positive integral exponents can occur. The ambient field is algebraically closed by the standard full-Hahn closedness theorem for algebraically closed characteristic-zero coefficients and a divisible value group; the repository also documents this theorem in its Hahn layer [5]. The fixed field is not algebraically closed: a square root of t would have valuation $1/2$, which is absent from $\mathbb{Z}$. Thus the fixed field in the cohomology calculation can lose algebraic closedness.

12.3 Compatibility is not the same as exactness

For any commuting family, compatibility only says that a tuple is a cocycle. The joint-resonant coefficients need not vanish. A simple additive example is a nonzero tuple of constants: it is automatically compatible, but no nonzero entry can occur in an exact tuple for a nonidentity rational dilation family.

A similarly simple multiplicative example is a tuple $(c_1, \dots, c_d) \in (\mathbf{k}^\times)^d$. It is compatible because the coefficient action is trivial. For a rational-dilation family with at least one nonidentity member, it is exact precisely when every $c_j = 1$. These examples realize every scalar obstruction appearing in Theorem 7.1.

13 Transfer to surreal and surcomplex normal forms

13.1 The normal-form dictionary

The standard Conway normal form writes a surreal as

$$x = \sum_{\beta < \alpha} r_\beta \omega^{a_\beta}, \quad r_\beta \in \mathbb{R}, \quad (a_\beta)_{\beta < \alpha} \text{ strictly decreasing,}$$

where α is an ordinal. The normal-form theorem, and its use as a Hahn-field representation, is described in [1]. With

$$t^g = \omega^{-g}, \tag{13.1}$$

the support is well ordered in increasing g -notation. Combining the supports of real and imaginary parts gives the corresponding complex-coefficient normal form for $\mathbf{No}(i) = \mathbf{No}(i)$.

Thus, for $q \in \mathbb{Q}_{>0}$, define the class field automorphisms

$$S_q \left(\sum_{\beta < \alpha} c_\beta \omega^{a_\beta} \right) = \sum_{\beta < \alpha} c_\beta \omega^{q a_\beta}, \quad c_\beta \in \mathbb{R} \text{ or } \mathbb{C}. \tag{13.2}$$

These are precisely the usual coefficient-fixing monomial dilations; their construction is not new. The new assertions concern what their difference equations and their centralizers determine.

13.2 Set-sized localization of every proof

A set of normal forms uses a set of exponents. For a finite family of class exponent automorphisms, close that set under their positive and negative integer iterates and under finite group operations. This closure is still a set. When division by rational scalars is needed, take its rational span. The resulting ordered subgroup supplies a set-sized Hahn field containing all input supports and all the orbit supports used in the additive constructions. Adding any exponent g needed for a multiplicative monomial equation still gives a set-sized subgroup. Neumann sums use finite sums of already included exponents.

More quantitatively, if the original union of supports has cardinal κ , these support-enlargement operations require at most $\max(\kappa, \aleph_0)$ exponents, apart from a finite number of explicitly adjoined monomial-solution exponents. This does not bound their surreal birthdays and does not assert that a chosen birthday fragment is closed under all these operations.

The ordinary cohomology groups in Theorems 5.2 and 7.1 remain statements about these set-sized fields. For the full proper-class fields, we state the corresponding *elementwise equation and formula assertions*. They involve set-supported series and finite tuples, not a purported set of all class cochains.

Theorem 13.1 (Surreal and surcomplex consequences). *Let $\mathbb{F}$ be $\mathbf{No}$ or $\mathbf{No}(i)$, with coefficient field $\mathbf{k} = \mathbb{R}$ or $\mathbb{C}$, respectively. Then the following assertions hold.*

1. *For $q \in \mathbb{Q}_{>0} \setminus \{1\}$, $S_q y - y = f$ is solvable in $\mathbb{F}$ exactly when the coefficient of ω^0 in f is zero. The solution is unique modulo $\mathbf{k}$.*

2. For $f \in \mathbb{F}^\times$, the equation $S_q y/y = f$ is solvable exactly when the leading normal-form coefficient of f is one. The solution is unique modulo $\mathbf{k}^\times$.
3. The simultaneous criteria of Theorem 7.1 hold for every finite family of positive rational dilations with at least one nonidentity member.
4. In the field expanded by S_q , the formulas in Sections 8 and 11 define the ordinary coefficient field, the Conway monomial class, every normal-form coefficient, the natural valuation ring, and all monomial truncations.
5. If a class field automorphism α commutes with S_q , then there are unique $\rho \in \text{Aut}(\mathbf{k})$ and an ordered additive class automorphism θ of $(\mathbf{No}, +, <)$ such that

$$\alpha \left(\sum_{\beta < \alpha_0} c_\beta \omega^{a_\beta} \right) = \sum_{\beta < \alpha_0} \rho(c_\beta) \omega^{\theta(a_\beta)}. \quad (13.3)$$

Conversely every such class map commutes with S_q . For $\mathbf{No}$, necessarily $\rho = \text{id}_{\mathbb{R}}$.

Proof. For the equation assertions, put the given supports into a set-sized Hahn workspace as just described and use the explicit solutions already proved. Necessity can also be checked directly on full normal forms; allowing larger supports does not remove a resonant coefficient obstruction.

The defining formulas hold for all surreal or surcomplex arguments, because their proofs use only the unique normal form of the element being tested and explicit set-supported witnesses. For the centralizer, repeat the preservation-of-definable-coefficients argument from Theorem 9.1. The normal-form monomial group is identified with $(\mathbf{No}, +)$ by (13.1), and the recovered valuation orientation gives its order. Additivity of θ makes the switch from g to $a = -g$ in (13.3) consistent. Every class image of a set support is again a set in a class theory with replacement for the class maps under consideration. Uniqueness of normal forms therefore gives the formula for each element. The converse is the standard strong monomial-lift calculation. Finally a field automorphism of $\mathbb{R}$ is the identity, as proved after Corollary 9.2. $\square$

Remark 13.2 (Foundational reading). These full-field statements can be read in NBG-style class mathematics, with each assertion about a class automorphism made for a specified class map. The notation for a centralizer of class automorphisms is structural shorthand, not an assertion that all such automorphisms form a set or even an ordinary class of set objects. All ordinary group cohomology and complete first-order theory claims in this article have set-sized versions and proofs. The same finite interpretation formulas also encode ordinary arithmetic and its power-set sort in the full surreal expansions; no global satisfaction class for arbitrary class structures is needed for those individual formula identities.

13.3 What is recovered, and what is not

On $\mathbf{No}(i)$, the theorem recovers the *complex* coefficient field and the canonical monomials. It does not claim that the distinguished real axis $\mathbf{No}$ is thereby definable. The centralizer still allows arbitrary coefficient-field automorphisms of $\mathbb{C}$; the ones preserving $\mathbb{R}$ are only the identity and conjugation. Thus preserving the recovered valuation structure is not the same as preserving a chosen real form.

Nor does S_q here mean the surreal exponential, the omega-map itself, a Berarducci–Mantova derivation, or an automorphism known to preserve any of those additional structures. The centralizer theorem is a result about an explicitly named *field* automorphism. It does not settle the exponential-automorphism image problems discussed in [1, 7]. Finally the full-surreal fine topology is not the summation mechanism used in the proofs. The repository distinguishes its small-net behavior from strong Hahn summation [6]; the present constructions stay on the latter side of that distinction.

14 A formalization route and precise limitations

14.1 A dependency-oriented route

A Lean development can be organized without attempting to formalize the model theory and the proper-class interpretation at the same time as the support algebra. The natural first target is a set-sized Hahn field, using the repository’s documented summability interface [5].

The first layer is order-theoretic: prove Lemma 3.1 as a summability certificate for the map $(n, s) \mapsto \theta^n s$ on the increasing chamber, and separately prove finite fibers. This should not be replaced by a convergence theorem. The labeled version then permits the construction of a strong linear Q_λ from the two iterate families and the fixed-exponent quotient.

The next layer establishes the two inverse identities, normalization, and the support/valuation bound. Once uniqueness of normalized solutions is available, the commutation lemma can be proved abstractly and need not repeat the coefficient calculation for every pair of automorphisms. The Koszul contraction is then finite exterior algebra with commuting linear maps. Its only Hahn-specific input is the previously verified resolvent package.

For multiplicative equations, the existing infinitesimal Hahn exponential/logarithm interface is the appropriate dependency. It is important that the development type the group U_1 separately from the whole multiplicative field, and keep the monomial-coordinate quotient as an explicit residual obstruction.

The definability theorem can first be recorded as algebraic equivalences: $S_2x = x^2$ iff x is a monomial; existence of y in $x - c = S_2y - y$ iff c is the constant coefficient; and the two geometric expansions deciding $B((1 - m)^{-1})$. A later model-theory layer can translate these identities into formula syntax. The unrestricted centralizer theorem then follows by preservation of these definable operations, rather than by assuming a strong automorphism instance at the outset.

14.2 Claims deliberately not made

There is no Lean source accompanying this manuscript and no claim of machine-checked infinite mathematics. The finite Python tests check signs, endpoint identities, and exterior-algebra contractions only. There is no uniform algorithm here for arbitrary Hahn names: deciding an exponent’s movement chamber or testing exact equality of coefficient values may itself require unavailable computational information.

The coefficient action in (2.1) is the identity. Allowing a nontrivial difference automorphism of $\mathbf{k}$ changes the fixed-exponent recurrences; division by $\chi(g) - \lambda$ is no longer the right operation. The finite family is commuting; no analogous contraction is asserted for a nonabelian acting group. The multiplicative results are abelian $K^\times$ results, not matrix gauge-classification theorems. No support-growth, convergence, or solvability theorem for arbitrary variable-coefficient Mahler systems is inferred from the scalar resolvent.

The use of full Hahn fields is substantive. A restricted support class must be checked for closure under increasing half-orbits, infinitesimal logarithms, and all relevant subseries before any theorem is transferred. Finally, characteristic zero is retained throughout the article. In characteristic two the monomial-recognition equation degenerates when coefficient squaring is trivial, so that particular formula cannot be silently exported.

14.3 Consolidated ledger of non-claims

The source states its limitations where they arise, and each is kept in place. The list collects them, with the place where each stands. Items 1–28 are the source’s; items 29–33 were added when the

report joined the collection.

1. Not refereed. No theorem is Lean-verified, no Lean source accompanies the report, and nothing is represented as already checked by Lean; the formalization route of Section 14 is a route, not a completed formalization (title page, Section 1, Section 14.2, Appendix C; the delivered README).
2. No named published open problem is claimed settled. In particular nothing here determines the existence or image of nontrivial exponential automorphisms of $\mathbf{No}$, and Camacho’s monomial-definability question in the smaller language without the extra automorphism is not answered (title page, Section 1, Section 13.3, Appendix C).
3. Priority is not certified. The main statements are proposed additions; the literature review was focused (Hahn automorphisms, Mahler solutions, valued difference fields, truncation definability), not exhaustive, and a later-located antecedent would change the priority assessment, not the proofs (title page, Section 1, Appendix C).
4. The repository was not independently built or exhaustively audited: only the catalogue, the root README and the surcomplex-automorphisms report’s guide were inspected, and that report’s full article was not audited (Section 1, bibliography entry [7]; the delivered README).
5. Credited, not claimed: monomial and character lifts (Kaplan–Krapp–Serra and the surcomplex-automorphisms report); the construction of the dilations on $\mathbf{No}$ and $\mathbf{No}(i)$; Hahn field operations and the normal-form dictionary; the support-to-arithmetic principle (Camacho); the principal-unit logarithm and exponential and the Neumann lemma (Sections 1, 6, 10, 13, Appendix C).
6. The rank-one Mahler identities of Examples 7.2 and 7.3 are illustrative calculations, not separate claims of novelty (Section 7).
7. The first-order, constant-scalar resolvent does not replace the higher-order, variable-coefficient theory of Faverjon–Roques; no support-growth, convergence or solvability theorem for arbitrary variable-coefficient Mahler systems is inferred (Section 1, Section 14.2).
8. No claim that these structures satisfy the residue difference closedness hypotheses of Pal’s relative quantifier elimination, and the coefficient-fixing structure is not identified with the existentially closed residue difference structures of other theories of valued difference fields (Section 1, the remark after Theorem 10.2).
9. The coefficient action is the identity; a nontrivial difference automorphism of $\mathbf{k}$ changes the fixed-exponent recurrences (Section 14.2, Section 15).
10. Finite commuting families only; no contraction is asserted for a nonabelian acting group (Section 14.2).
11. The multiplicative results concern the abelian group $K^\times$, not matrix gauge classification; no universal smaller formula for $H^*(\mathbb{Z}^d, M)$ is claimed for arbitrary shears and characters (Sections 6 and 14.2).
12. The resonance projection is linear but not a field homomorphism, and $[x]_0$ is not a ring homomorphism on K (Section 2, the remark after Corollary 4.3).
13. The kernel statements do not say that K is a direct sum of eigenspaces (after Corollary 4.2).
14. The Koszul complex is a complex of additive groups; its naive coefficientwise wedge product is not claimed to make d an ordinary derivation, and Δ_j satisfies no ordinary Leibniz rule (Section 5).

15. Lemma 3.1 is not an assertion of cofinal growth; strong summation is not valuation convergence of finite partial sums, and no support proof is replaced by “the orbit tends to infinity” (Sections 2 and 3, and the shear example (12.1)).
16. The single-equation criteria (7.4) and (7.5) are existence equivalences, not claims that the equations are solvable for every right-hand side (Section 7).
17. Full Hahn fields are essential: restricted support classes need separate closure checks, and definability in the full field supplies no algorithm for arbitrary presentations (the remark on fullness in Section 8, Section 14.2, Section 15).
18. Characteristic zero is retained; the monomial-recognition formula is not exported to characteristic two (Section 14.2).
19. The undecidability theorem concerns the complete theory of the specified structure; it asserts no failure of decidability for every smaller language of a valued field, and interpretation with a named parameter alone would not justify the reduct claim (Section 10).
20. The interpreted arithmetic does not follow merely from naming a valuation ring; adding S_2 is a genuine expansion, so there is no contradiction with tame valued-field reducts (the remark after Theorem 10.2).
21. Interdefinability of the rational dilations does not imply conjugacy (Section 11).
22. Ordinary group cohomology and complete first-order theories are set-sized only. There is no set of class automorphisms or class cochains; the centralizer of class automorphisms is structural shorthand; the support localizations carry no birthday bound, and no birthday fragment is asserted closed under them (Section 13).
23. On $\mathbf{No}(i)$ the real axis $\mathbf{No}$ is not recovered, and the centralizer contains all automorphisms of $\mathbb{C}$; preserving the recovered valuation structure is not preserving a chosen real form (Section 13.3).
24. S_q is not the surreal exponential, the omega-map, a Berarducci–Mantova derivation or an automorphism known to preserve any of these; that strong coefficient-fixing automorphisms commute with the infinitesimal Hahn exponential is not a claim that a monomial automorphism commutes with the global surreal exponential (Sections 6 and 13.3).
25. The fine topology of the full surreal field is not the summation mechanism of any proof (Section 13.3).
26. No uniform algorithm for arbitrary Hahn names is supplied; deciding a movement chamber or testing coefficient equality may need unavailable information (Section 14.2).
27. The questions of Section 15 are raised here, not claimed to have been posed before in exactly this form, and the boundary of the method they describe is not a claim that the general problem is impossible (Section 15).
28. The 66,131 finite checks are not a formal verification: they do not decide Hahn summability, quantify over ordered groups, check the first-order interpretations or verify proper-class assertions; compiling and inspecting the PDF and running the script were not independent review (title page, Appendices A and C; the delivered README and PROOF_AUDIT.md).
29. Added: none of the four questions of the surcomplex-automorphisms report (saut:sec:questions) is answered. Its non-claims that the four-layer decomposition covers G_v only and that $\text{Aut}(\mathbf{No}(i))$ is not classified are extended only by the one subgroup $\text{Cen}(S_q)$, which is shown to lie in G_v ; the unrestricted group is not classified (Section 1.5).
30. Added: the rigidity report’s Question 13.1 (q:image) is untouched; nothing here says which canonical lifts $\hat{T}$ commute with the exponential (Section 1.5).

31. Added: the dilations here are not the argument dilations $f(z) \mapsto f(\lambda z)$ of the holonomic-rigidity report, and no statement of that report is used or extended (Section 1.5).
32. Added: the arithmetic interpretation and undecidability of the birthday-cutoffs report concern a different expansion of **No**; neither result implies the other (Section 1.5).
33. Added: no `dsup` label has a Lean mapping in docs/FORMALIZATION.md; the Lean Hahn-series interfaces the source cites (strong summation, the Neumann lemma, infinitesimal exponential and logarithm, algebraic closedness) prove none of this report's statements (Section 1.5, Appendix D.2).

15 Further questions suggested by the results

The following are questions raised by this article, not problems claimed to have been previously posed in exactly this form.

Question 15.1 (Minimal closure for definable normal forms). Characterize S_2 -invariant subfields $L \subseteq \mathbf{k}((t^\Gamma))$ on which $\text{im}(S_2 - I) = \{f \in L : [f]_0 = 0\}$ and every required coefficient shift x/t^g remains available. Which natural restricted-series fields recover truncation without containing all full Hahn subseries?

The proof shows what must be tested: the two half-orbit sums supply additive witnesses, while truncation additionally requires the relevant subseries to exist in L . Those are different closure requirements. Failure of one need not imply failure of the other.

Question 15.2 (Nontrivial coefficient difference structure). Let $\rho \in \text{Aut}(\mathbf{k})$ and $\sigma(\sum a_g t^g) = \sum \rho(a_g) \chi(g) t^{\theta g}$. Can one obtain an equally explicit decomposition in which each fixed-exponent coefficient is replaced by the kernel and cokernel of a scalar-twisted operator on $(\mathbf{k}, \rho)$?

The increasing-orbit support lemma still applies. The obstacle is not support transport but the residual coefficient equations and their compatibility under several automorphisms. A satisfactory result would separate those two mechanisms rather than impose residue difference closedness as an unexplained global substitute.

Question 15.3 (Definability from other exponent automorphisms). Which ordered exponent automorphisms θ admit a finite collection of polynomial equations in $x, \sigma x, \dots$ that recognizes canonical monomials? Can the centralizer argument be extended to shears with a large fixed Hahn field, where the definable projection lands in that field rather than in $\mathbf{k}$?

For rational dilations, the power equation and the vanishing of the relevant scalar eigenspace solve both tasks. A shear lacks that simple power relation; its fixed-field projection alone does not distinguish ordinary coefficients from all fixed series. This is a concrete boundary of the method, not a claim that the general problem is impossible.

16 Conclusion

The article begins with a support calculation and ends with a structural and logical reconstruction theorem. The support calculation has no rank-one hypothesis: moving exponent orbits have summable increasing halves, and only pointwise-fixed resonant exponents obstruct scalar difference equations. For commuting actions, the normalized inverses combine into an explicit contraction in every cohomological degree. The multiplicative problem then separates into a coupled exponent/leading-coefficient condition and an exactly computable logarithmic obstruction.

For doubling, the resonant projection is just the ordinary constant coefficient. Together with the equation $S_2x = x^2$, it recovers the entire normal form inside the field language with one automorphism. This forces all centralizing field automorphisms to be strong untwisted lifts, and it makes full arithmetic support coding available. In this precise sense, a single elementary symmetry both rigidifies field automorphisms and greatly increases the information definable in the field. The same proofs yield explicit new statements for surreal and surcomplex normal forms without confusing proper-class topology, formal Hahn summation, or the separate exponential structure.

A Endpoint identities and finite verification

The accompanying `verify.py` uses only Python's standard library and exact rational arithmetic. Its output is recorded in `verification.txt`. (As placed in the collection these are `code/verify.py` and `data/verification.txt`; Appendix D.3 gives the rerun.) It is intentionally a collection of finite checks rather than a finite approximation advertised as proof of an infinite statement.

For a truncation of the increasing-chamber part, define

$$Q_{\lambda,N}^+ f = - \sum_{n=0}^{N-1} \lambda^{-n-1} \sigma^n f.$$

A finite telescope gives

$$(\sigma - \lambda) Q_{\lambda,N}^+ f = f - \lambda^{-N} \sigma^N f. \quad (\text{A.1})$$

For the decreasing chamber, define

$$Q_{\lambda,N}^- f = \sum_{n=1}^N \lambda^{n-1} \sigma^{-n} f.$$

Then

$$(\sigma - \lambda) Q_{\lambda,N}^- f = f - \lambda^N \sigma^{-N} f. \quad (\text{A.2})$$

The program tests these identities on rational exponents under doubling and on integer-coordinate exponents under shears, including nontrivial characters. It retains the endpoint term exactly rather than treating it as a small numerical error.

The program also checks the Koszul identities on finite diagonal coefficient models in every exterior degree for $d = 1, \dots, 5$. These models exercise the signs of ε_j, ι_j and the ordered factors E_{j-1} , including coordinate directions with zero difference operator. It checks both the homotopy identity and the annihilation/normalization identities on basis vectors. This is a check of the finite operator algebra appearing in the proof, not a reduction of arbitrary Hahn operators to finite diagonal matrices.

Additional checks compare the first finitely many coefficients in the multiplicative product of Example 7.3, test explicit phase-twist resonance projections, and demonstrate the failure of a nontrivial character twist to commute with doubling on a monomial. The program does not decide Hahn summability, quantify over all ordered groups, check the first-order interpretations, or verify proper-class assertions. Those tasks remain in the written proofs.

B A compact formula dictionary

This dictionary uses S_2 and the definable functions whose existence and uniqueness were proved above. Every occurrence of a function can be eliminated in favor of its graph formula.

Recovered object	Definition or test
Coefficient field	$C(c) \iff S_2c = c.$
Constant coefficient	$B(x) = c \iff C(c) \wedge \exists y (x - c = S_2y - y).$
Canonical monomial	$M(m) \iff m \neq 0 \wedge S_2m = m^2.$
Coefficient at m	$\text{coef}(x, m) = B(x/m), \text{ for } M(m).$
Support	$\text{Supp}(m, x) \iff M(m) \wedge \text{coef}(x, m) \neq 0.$
Positive exponent	$J(m) \iff M(m) \wedge m \neq 1 \wedge B((1 - m)^{-1}) = 1.$
Exponent comparison	$v(m) < v(n) \iff J(n/m), \text{ for monomials.}$
Valuation ring	Every monomial in the support is 1 or satisfies J .
Infinitesimal ideal	Every monomial in the support satisfies J .
Leading monomial	The unique support monomial of least recovered exponent.
Truncation at m	Preserve the coefficient at n exactly when $J(m/n)$; set it to zero otherwise.
Natural-number sort	$N_m = \{u : \text{Supp}(u, (1 - m)^{-1})\}, \text{ for } J(m).$
Power-set sort	Series with coefficients in $\{0, 1\}$ and support contained in N_m .

The monomial 1 has exponent zero, not natural-number index one. In the interpreted natural-number sort the field element $1 = m^0$ represents $0 \in \mathbb{N}$, whereas the parameter m represents $1 \in \mathbb{N}$. This distinction is essential when translating arithmetic formulas.

C Provenance, reproducibility, and novelty ledger

The delivery contained this standalone L^AT_EX source, its compiled PDF, a standard-library verification script, the exact output of that script, a README, and a small Makefile. The bibliography is internal, so no Bib_TE_X database is required. In the delivered flat layout a typical build was

```
python3 verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The PDF was compiled from the delivered source and visually inspected through rendered page images. The finite test output was generated by running the delivered script. Neither activity is described as an independent mathematical review. (As placed in the collection the delivered PDF and README are not shipped; the files and working commands are in Appendix D.3.)

Result or ingredient	Status in this manuscript
Hahn field operations; normal forms	Standard background, with the normal-form dictionary cited to the literature.
Monomial and character lifts	Prior construction; explicitly credited to the literature and repository.
Orbit-resolvent theorem	Proved here for arbitrary ordered exponent automorphisms; proposed addition, priority not certified.
Full Koszul contraction	Proved here with an explicit strong homotopy; proposed addition.
Principal-unit logarithm	Standard formal Hahn construction; used, not claimed new.

Result or ingredient	Status in this manuscript
Multiplicative obstruction theorem	Proved here from the contraction and multiplicative coordinates; proposed addition in this general form.
Single-dilation recovery	Main proposed new structural result; explicit formulas and proofs supplied.
Unrestricted centralizer	Main proposed new algebraic consequence; no valuation or strength hypothesis assumed.
Support-to-arithmetic principle	Prior precedent in Camacho; a Boolean coding and uniform parameter elimination are provided in the present expansion.
Rational interdefinability and nonconjugacy	Proved consequences with distinct meanings; not a claim about exponential automorphisms.
Lean coverage	None claimed for these new theorems. A dependency route, not a completed formalization, is supplied.
Published open problems	No named existing open problem is claimed settled; limitations are explicit.

The bibliography review was focused on Hahn automorphisms, Mahler solutions, valued difference fields, and truncation definability. It was not an exhaustive search of all unpublished notes or every language’s literature. A subsequently located antecedent would change the priority assessment, not the logical content of the proofs. The intended research contribution is the collection of exact statements and their deductions, with the single-dilation recovery theorem as its organizing result.

D Provenance of this report

D.1 One manuscript

This report is built from *one* manuscript: the archive `surreal_dilation_support`, manuscript 09 of batch 22 of the collection, dated 22 September 2026 and pinned to repository commit 465a54b479a1ee842cbf7db1689a7d2f6bfe25e1. It was placed in commit 7b5f934 as a new report: it answers no question that a report of the collection names, and it continues only one non-claim of the surcomplex-automorphisms report (Section 1.5). There was no second source; nothing was merged and nothing was selected away. Every result, proof, example, question and limitation of the manuscript is printed here.

The delivered verification program code/`verify.py`, its Makefile code/`Makefile`, the recorded output data/`verification.txt` and the author-side audit `PROOF_AUDIT.md` are kept byte-identical. The audit and the Makefile describe the delivered flat layout (see Appendix D.3). The manuscript’s article and README were rewritten, and the delivered PDF was replaced by a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `dsup:`. The source’s 99 labels are kept, unchanged after the prefix, and nine were added (`dsup:sec:conventions`, `dsup:sec:position`, `dsup:sub:recovered`, `dsup:sub:notmade`, `dsup:sec:nonclaims`, `dsup:app:report`, `dsup:app:onesource`, `dsup:app:pinned`, `dsup:app:files`).
- (2) The exponent dilation D_q is written S_q , the abbreviation D for D_2 is dropped, and the support formula $S(m, x)$ is written $\text{Supp}(m, x)$ (Section 1.4). The sentence introducing the language at the start of Section 8 and the first sentence of Appendix B were adjusted accordingly.

- (3) The title page names the report as a single-source report of the collection and extends the status paragraph. Sections 1.4 and 1.5, the ledger of Section 14.3 (with five items added) and this appendix are new. Short sentences pointing to them, and to the files as placed, were added at the end of Section 1 and in Appendices A and C; the latter's description of the delivery is put in the past tense.
- (4) The columns of the two appendix tables are set ragged right, which removes two underfull lines of the delivered build.

No mathematical statement was changed. Where the edit had to choose: the dilation was renamed rather than keeping D_q , because the notation guide reserves D for a scalar derivation and the nearest report already calls the map S_a ; the support formula was renamed rather than giving the dilation a third letter; the title is kept; and the report is placed among the surcomplex reports, because its results are uniform in the coefficient field, Theorem 13.1 treats **No** and **No**(i) together, and its nearest neighbour is the surcomplex-automorphisms report.

D.2 Corrections since the pin

The source compared itself with the collection at its pin. Checked against the collection at the placement commit 7b5f934, its repository statements stand; one addition matters.

- The root README still documents strong Hahn summation, the full Neumann support lemma, coefficient maps preserving strong sums, the infinitesimal exponential and logarithm, and algebraic closedness of Hahn fields over algebraically closed characteristic-zero coefficients; these are Lean statements of the collection's Hahn layer, and none of them is a statement of this report. The catalogue still separates the small-net behaviour of the fine topology from strong Hahn summation. The surcomplex-automorphisms report still constructs the coefficient, exponent and character lifts (saut:thm:monomial); neither its article nor that of the rigidity report has changed since the pin.
- At the pin the catalogue listed 44 research reports in five families. Batch 21, placed in d4e71b7 after the pin, added two, among them the birthday-cutoffs report whose undecidability results are compared in Section 1.5; batch 22, placed in 7b5f934, adds further reports, this one among them.
- The surcomplex-automorphisms guide that the source inspected gained, after the pin, one paragraph on the two batch-21 reports that use conjugation on **No**[i]; it bears on no statement here. The source did not audit that report's article; the relations of Section 1.5 were checked against the article itself.

D.3 Files as placed and rerun

File	Role
code/verify.py	the delivered finite checks (delivered as <code>verify.py</code>)
code/Makefile	the delivered Makefile, for the flat delivered layout
data/verification.txt	the delivered recorded output
PROOF_AUDIT.md	the delivered author-side audit

The program uses only the Python standard library, prints to standard output and writes no file. The Makefile, however, assumes that `article.tex`, `verify.py` and `verification.txt` sit in one directory: its target `check` runs `python3 verify.py > verification.txt`, overwriting the recorded output in place, and `all` also rebuilds `article.pdf`. It must not be run in this directory. Two working reruns, from the report directory, with a fresh scratch directory `$T`:

```
python code/verify.py > $T/direct.txt
diff --strip-trailing-cr $T/direct.txt data/verification.txt

cp code/verify.py code/Makefile $T/ && make -C $T check
diff --strip-trailing-cr $T/verification.txt data/verification.txt
```

Both were run for this report under Python 3.14.4 on Windows (the second with GNU Make). Each printed 17 PASS groups and 66,131 checks and reproduced data/verification.txt exactly apart from line endings (Windows writes CRLF).

References

- [1] Elliot Kaplan, Lothar Sebastian Krapp, and Michele Serra, *Decomposing the automorphism group of the surreal numbers*, arXiv:2509.22374, version 3, 23 April 2026. <https://arxiv.org/abs/2509.22374v3>. The normal-form and monomial-automorphism preliminaries, and the distinction from exponential-preserving automorphisms, are the relevant background.
- [2] Santiago Camacho, *Truncation in Hahn Fields is Undecidable and Wild*, arXiv:1706.03722, version 1, 12 June 2017. <https://arxiv.org/abs/1706.03722v1>. Sections 4–6 provide the arithmetic-interpretation and support-definability precedent. The present article adds a distinguished dilation instead of assuming the same truncation signature.
- [3] Colin Faverjon and Julien Roques, *Hahn series and Mahler equations: algorithmic aspects*, Journal of the London Mathematical Society **110** (2024), e12945. <https://doi.org/10.1112/jlms.12945>. Accepted manuscript: arXiv:2412.04928v1, <https://arxiv.org/abs/2412.04928v1>.
- [4] Koushik Pal, *Multiplicative Valued Difference Fields*, arXiv:1011.1655, version 2, 13 November 2010. <https://arxiv.org/abs/1011.1655v2>.
- [5] Vladimir Reshetnikov, *Surreal* repository, root README and documented Hahn-series formalization interfaces, snapshot 465a54b479a1ee842cbf7db1689a7d2f6bfe25e1, reviewed 22 September 2026. <https://github.com/VladimirReshetnikov/Surreal/tree/465a54b479a1ee842cbf7db1689a7d2f6bfe25e1>.
- [6] Vladimir Reshetnikov, *The surreal and surcomplex reports*, docs/README.md in the same repository snapshot. AI-assisted research-report catalogue and statement of verification scope. <https://github.com/VladimirReshetnikov/Surreal/blob/465a54b479a1ee842cbf7db1689a7d2f6bfe25e1/docs/README.md>.
- [7] *Automorphisms of the Surcomplex Numbers: Real forms, Hahn symmetries, valuation, and rigidity*, research report prepared with ChatGPT, 22 September 2026; repository guide docs/surcomplex/surcomplex-field-automorphisms/README.md, in the same snapshot. The guide’s statement inventory was inspected; its complete article was not independently audited here. <https://github.com/VladimirReshetnikov/Surreal/tree/465a54b479a1ee842cbf7db1689a7d2f6bfe25e1/docs/surcomplex/surcomplex-field-automorphisms>.